STATISTICS 475: Applied Discrete Data Analysis

Lecture 13: Generalized Linear Models

(B&L Section 2.3)

1 Problem to be solved

- You now know a fair amount about two very important forms of regression model: Linear and Logistic
- While they have distinctly different uses and properties, they also have similarities in structure
  - Both start with a distribution: normal or binomial
  - Both focus on modeling mean parameters (π is the expected value [mean] of a Bernoulli RV)
  - Both combine explanatory variables and parameters into linear combination (the linear predictor)
  - Both have extensions for transformations, interactions, categorical explanatories
- They also have similarities in inferences
  - Wald tests are similar to t-tests and CIs
  - LR tests are similar to F-tests that compare two models
  - Residual Deviance is similar to Sum of Squared Errors, SSE (also called "Residual Sum of Squares", RSS)
- Is this a coincidence?
  - Can we develop other models with similar structure?

2 Generalized Linear Models

The first thing we have to do is to define a "model" more clearly:

- In statistics, a "model" generally refers to the *entire distributional structure*
  - $Y \sim N(\mu, \sigma^2)$,where $\mu = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p$
  - $Y \sim Bin(n, \pi)$,where $\text{logit}(\pi) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p$
  - There are many other distributions for different data structures or relationships
- Notice the similarities in the two models above:
  1. Each has a specific distribution
  2. Each has a parameter representing the mean
  3. Some function of each mean is assumed to be linearly related to the explanatory variables and some regression parameters
- Both of these are examples of a class of models called GENERALIZED LINEAR MODELS (GLMs)

2.1 Definition of a generalized linear model

- A GLM consists of the following three components
  - The RANDOM COMPONENT is the particular distribution (or "distributional family") being used to describe the data.
    - Normal
    - Binomial
      - Bernoulli=$Bin(1, \pi)$
    - We will learn others
      - In particular, all of these distributions are members of the EXPONENTIAL FAMILY OF DISTRIBUTIONS (see appendix, if you dare...)
    - Each random component has a parameter that we want to relate to explanatory variables
  - The SYSTEMATIC COMPONENT is the mathematical form used to combine the explanatory variables with the regression parameters.
    - In both linear and logistic models, and in all other GLMs, we use a *linear* systematic component, $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p$
      - Called the LINEAR PREDICTOR
    - Other (nonlinear) forms of systematic components are possible
      - Generalized *nonlinear* models
  - The LINK FUNCTION $g(\cdot)$ specifies how the modeled parameter of the random component is connected to the systematic component

    - In other words, it's the function of the mean parameter that we equate to the linear predictor.
  - If θ represents a model's mean parameter, then the link has the form

$$g(\theta) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p,$$

    - In the normal random component,

$$g(\mu) = \mu = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p,$$

      so this is called the *IDENTITY LINK*
    - In the binomial random component,

$$g(\pi) = \log(\frac{\pi}{1-\pi}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p,$$

      so this is called the LOGIT LINK
      - Recall `glm(...,family=binomial(link=logit),...)`.
      - *You can use other link functions here to explore different shapes*
    - We have to invert the link (solve it for θ) to get back the mean parameter as a function of the linear predictor
      - Inverting the identity is easy—it's just itself
      - Inverting the logit gives the familiar form $\pi = \exp(\cdot)/[1 + \exp(\cdot)]$
      - Other links have other inverses.
    - Different links lead to different shapes in the relationship between the modeled parameter and the explanatories
    - Different links also impose bounds on those parameters as linear predictor goes toward ∞ or $-\infty$
    - There are also certain mathematical reasons to use certain links (simplifies likelihoods, makes inferences easier to construct, asymptotic approximations can be better.

3 Notes

- The term "generalize" means "to make more widely applicable"
  - GLMs are generalizations to the normal linear regression model that allow it to be extended to other distributions
    - (In R, `lm()` fits **l**inear **m**odels, and `glm()` fits **g**eneralized **l**inear **m**odels, including the normal one!)
- The term "logistic regression" refers specifically to using the logit link with binomial data.
  - In the exercise, you will explore other links.

  - The math behind computing ORs is much simpler for the logit link (which is defined from odds!) than for other possible links, making it more appealing when it gives a reasonable fit to the data.

- We will learn additional GLMs
  - Chapter 3: multinomial distribution with various links
  - Chapter 4: Poisson distribution with log link

4 Appendix: Exponential Family

Actuarial exams may ask questions about the exponential family of distributions, so for the Act Sci students and other cool math nerds, here's a little bit more on that concept. This summary is from the book *Generalized Linear Models* by McCullagh and Nelder, which has WAY more details if you are interested. No one needs to know this for STAT 475, and I don't plan to spend time going through the algebra with students. This is strictly FYI for those who need it.

- Generalized linear models can technically be created for a response variable Y from any probability distribution $F(\theta)$ where θ is the parameter for the mean of F.
- Some distributions have forms that make it easy to prove theory relating to, e.g., finding MLEs and their sampling distributions.
- One such family is the EXPONENTIAL FAMILY OF DISTRIBUTIONS,
  - Probability densities or masses are members of this family if they can be rearranged into the form

$$f(y;\theta,\phi) = \exp[(y\theta - b(\theta))/a(\phi) + c(y,\phi)] \tag{1}$$

    * θ is the "canonical parameter" that y\muis typically the target of inference
      · Mean μ in normal; probability π in binomial
    * ϕ is an auxiliary parameter, often a dispersion parameter in models that have more than one parameter
      · Variance σ^2 in normal; not present in binomial
    * $a(\cdot), b(\cdot), c(\cdot)$ are three functions that are specific to the distribution
      · Whatever they need to be to allow the distribution to be written in this way
      · Notably, $b()$ is a function only of the canonical parameter θ and c is a function of y and ϕ,but must not contain θ
      · θ may appear only in the numerator of the first term and must be expressible as shown.
    * $E(Y) = \frac{\partial b(\theta)}{\partial \theta}$
    * $Var(Y) = \frac{\partial^2 b(\theta)}{\partial \theta^2} a(\phi)$
  - Example: Normal density $f(y;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp((y-\mu)^2/2\sigma^2)$
    * Can be written in the form $= \exp[(y\mu - \mu^2/2)/\sigma^2 - \frac{1}{2}(y^2/\sigma^2 + \log(2\pi\sigma^2))]$
      · $y\theta = y\mu$ so $\theta = \mu$
      · $b(\theta) = \theta^2/2$
      · $a(\phi) = \sigma^2$, so can set $\phi = \sigma^2$ and $a(\phi) = \phi$
      · $c(y,\phi) = \frac{1}{2}(y^2/\phi + \log(2\pi\phi))$
  - Example: Binomial (try it!)

- If a distribution is a member of this family, then there is theory that promises certain results:
  - MLEs exist (likelihood has a proper maximum)
  - "Regularity" conditions for asymptotic properties of MLEs are automatically satisfied.
  - Asymptotic approximations work better for smaller sample sizes than otherwise
  - Mathematics are simpler when the "canonical link" is used (logit for binomial, identity for normal)

5 Exercises (due on date to be announced)

Compete the following exercises. As always, add proper interpretations on all confidence intervals and tests.

1. Read the rest of Section 2.3 on links for binary regression models and complete the following for the Professional Dues data using only the proposed dues increase as an explanatory variable as we did in Lecture 7.

   (a) Fit and report the model using a logit link (just repeating the Lecture 7 fit).

   (b) Fit and report the model using a probit link (`link="probit"`)

   (c) Fit and report the model using a complementary log-log link (`link="cloglog"`)

   (d) Plot the probability estimates from these three models against the dues increase on one graph, using three different colours. Add a legend if you know how, or explain in text which colour is for which model.

   (e) For each model, compute the probability of that someone quits using dues increases of \$30, \$40, and \$50 and find corresponding confidence intervals.

   (f) Compare the results from these different analyses: are they similar or are there specific ways in which they give somewhat different results?

   (In Chapter 5 we will learn how to choose which model might be best.)

STATISTICS 475: Applied Discrete Data Analysis

Lecture 14: The Multinomial Distribution

(B&L Section 3.1)

1 Problem to be solved

- Logistic regression gives a model only for binary response variables
- Many categorical measurements have more than two possible outcomes
  - How can we model those?

2 Multinomial Probability Distribution

- In the binomial distribution:
  - We had an RV Y that was binary
  - Focused on calling one category "success" and modeling its counts from n Bernoulli trials
  - Unique situation where knowing what *didn't* happen is the same as knowing what *did*
    - If not success, then failure
    - Probability of success is π; probability of failure is implicitly $1 - \pi$
    - $W \sim Bin(n, \pi)$, the count of successes in n trials (n is fixed):

$$P(W = w) = \frac{n!}{w!(n-w)!}\pi^w(1-\pi)^{n-w}$$

      - $n - w$ is automatically the count in the other category
      - Distribution is the same if I model the $n - w$ failures from $Bin(n, 1 - \pi)$

- Consider an alternative set of notation for the same thing, where we ignore this feature and think of categories as equally important:
  - Instead of modeling one count RV W, model two count RVs, N_1 and N_2
    - No longer "Success" and implicit "Failure" as simply lack of success
    - Instead, think of "Outcome 1" and "Outcome 2"
  - Instead of observing w and computing $n - w$, observe two counts n_1 and n_2
    - Require $n_1 + n_2 = n$
  - Instead of one parameter for the probability of success, have one for each category, π_1 and π_2
    - Require $\pi_1 + \pi_2 = 1$
  - Our binomial PMF can be rewritten equivalently as a joint distribution of N_1 and N_2:
    $$P(N_1 = n_1, N_2 = n_2) = \frac{n!}{n_1! n_2!} \pi_1^{n_1} \pi_2^{n_2},$$
    subject to $\pi_1 + \pi_2 = 1$ and $n_1 + n_2 = n$
- Now consider a more general categorical variable, Y, with outcomes $1, 2, \ldots, J$
  - Summarize as counts in each category
    - RVs $N_1, N_2, \ldots, N_J$
    - Probabilities of each category $\pi_1, \pi_2, \ldots, \pi_J$
    - Observed counts $n_1, n_2, \ldots, n_J$
  - The joint probability distribution for the whole set of counts is
    $$P(N_1 = n_1, N_2 = n_2, \ldots, N_J = n_J) = \frac{n!}{n_1! n_2! \ldots n_J!} \pi_1^{n_1} \pi_2^{n_2} \ldots \pi_J^{n_J},$$
    subject to $\sum_{j=1}^{J} \pi_j = 1$ and $\sum_{j=1}^{J} n_j = n$
    - Some sources replace π_J with $1 - \sum_{j=1}^{J-1} \pi_j$ and n_J with $n - \sum_{j=1}^{J-1} n_j$
  - This is the MULTINOMIAL DISTRIBUTION

Properties of the multinomial distribution

- Like in binomial, for each $j = 1, \ldots, J$:
  - Expected count in each category is $E(N_j) = n\pi_j$
  - Variance of each count is $Var(N_j) = n\pi_j(1 - \pi_j)$
- Covariance between counts in categories a and b is $Cov(N_a, N_b) = -n\pi_a\pi_b$
  - Negative, because the total count n is fixed, so every time one count grows, it is one less potential count in the rest

Estimation and Inference

- Like in the binomial, ML estimation is used to estimate each π_j
  - The likelihood is same as the PMF
  - MLEs are sample proportions, $\hat{\pi}_j = n_j/n$, $j = 1, \ldots, J$
- Wald and LR inferences apply (as well as Score when possible)
  - Need larger overall n to spread adequately large counts across all categories
- *Degrees of freedom will usually need to be calculated.*

3 Uses of the Multinomial Distribution

1. Model for a contingency table (see next lecture)
2. Model for regression of multicategory responses on explanatory variables (complicated: we will only touch on this)
   (a) How do various explanatory variables relate to the probability that someone during the pandemic was unvaccinated, partially vaccinated, fully vaccinated, or vax+boosted against Covid?
3. Model for Pearson (chi-squared) goodness of fit test
   - "Goodness of fit" is a generic concept measuring how well a model fits data
   - Simplest (and rarest) application is this:
     - Suppose you have reason to think that the probabilities in the categories are specifically $\pi_1^0, \pi_2^0, \ldots, \pi_J^0$.
     - We can test whether data fit these probabilities
       * $H_0 : \pi_1 = \pi_1^0, \pi_2 = \pi_2^0, \ldots, \pi_J = \pi_J^0$ versus $H_a :$ no restriction on the π_j's
         · When H_0 is true, the expected counts in the groups should be $n\pi_1^0, n\pi_2^0, \ldots, n\pi_J^0$
         · To test this null, we measure whether the observed counts lie "too far" (in probability terms) from this pattern.
     - The score test statistic is known as the Pearson Statistic, comparing observed counts to what is expected when H_0 is true:

$$X^2 = \sum_{j=1}^{J} \frac{(n_j - n\pi_j^0)^2}{n\pi_j^0}$$

     - Under H_0, X^2 has approximately a χ^2_{J-1} distribution (why $J-1$?) in "large" samples (see Section 4 for details).
       * What is "large"?
         · All $n\pi_j^0 \geq 5$?
         · All $n\pi_j^0 \geq 1$ and at least 1/4 of $n\pi_j^0 \geq 5$?
         · It varies! See Appendix!!!

Example: Hybrid Crosses (Lecture 14 scripts.R)

In genetics, two traits are inherited independently if the probabilities with which their combinations appear in a certain type of experiment are 9/16, 3/16, 3/16, and 1/16, abbreviated as 9:3:3:1. Therefore, a test for independence of the traits can be performed by repeating this experiment many times, counting the number of results in each of 4 groups, and testing whether the model of a 9:3:3:1 ratio fits the observed data.

In one study of tomatoes, two traits were being studied: stature (height: tall/dwarf), and leaf structure (cut/potato). To see whether these two traits are inherited independently of each other, 1611 plants were cross-bred and their trait combinations measured. The counts were 926 "Tall, cut-leaf", 288 "Tall, potato-leaf", 293 "Dwarf cut-leaf", and 104 "Dwarf potato-leaf".[1]

The random variable Y here is the observed combination of traits and it takes on four possible outcomes with probabilities denoted by $\pi_1 = P(\text{Tall, cut-leaf})$, $\pi_2 = P(\text{Tall, potato-leaf})$, $\pi_3 = P(\text{Dwarf cut-leaf})$, $\pi_4 = P(\text{Dwarf potato-leaf})$. We test the null hypothesis that the probabilities follow the 9:3:3:1 ratio against the alternative that they don't. That is, we have

$$H_0 : \pi_1 = 9/16,\ \pi_2 = 3/16,\ \pi_3 = 3/16,\ \pi_4 = 1/16.$$

The corresponding expected counts assuming the 9:3:3:1 ratio are $n\pi_j^0$: 1611*9/16=906.1875; 1611*3/16=302.0625; and 1611*1/16=100.6875. Plugging these into the Pearson statistic yields

$$X^2 = \sum_{j=1}^{J} \frac{(n_j - n\pi_j^0)^2}{n\pi_j^0} = \frac{(926 - 906.19)^2}{906.19} + \ldots + \frac{(104 - 100.69)^2}{100.69} = 1.47$$

with a p-value of 0.69 on χ^2_3. Since $0.69 > 0.05 = \alpha$, we do not reject the null hypothesis. We conclude that there is not sufficient evidence that the probabilities deviate from this pattern. That is, it appears that the two traits *may be* inherited independently of each other.

This test can be conducted in R in one line:

```
chisq.test(x=c(926, 288, 293, 104), p=c(9,3,3,1)/16, correct=FALSE)
```

---

4 Notes

- Pearson test is a score test
  - Preferred type of test
  - A LR test can be used instead of the Pearson test, but not expected to be better.

[1]Source: Sokal, R.R., and Rohlf, F.J. (1995) *Biometry: The Principles and Practice of Statistics in Biological Research, 3rd Ed.* New York: W.H. Freeman, p.692. See also http://living-mudflower.blogspot.com/2015/02/tomatoes-potato-leaf-vs-regular-leaf.html for an easy description of leaf types.

- Pearson test (like all score tests) is an asymptotic test, meaning that χ^2 holds approximately, and the approximation is better in large samples
  - Usual advice is all $n\pi_j^0 \geq 5$, but this is a soft guideline.
    - I have also seen "All $n\pi_j^0 \geq 1$ and at least 1/4 of $n\pi_j^0 \geq 5$"
  - Appendix and Lecture 14 Script has a simulation where you can play with the sample size and see how the actual distribution of X^2 compares to χ_3^2 in our example.

5 What to learn from this

- Multinomial is J-category extension of binomial
  - Form of density function is the same
  - Estimation of π_j's by ML and inferences by score, LR, and Wald as available
- Pearson tests are used all over the place
  - General form is $\sum \frac{(\text{observed}-\text{expected})^2}{\text{expected}}$
    - Expected is computed under whatever null hypothesis is being tested
    - Approximate χ^2 distributions, with DF depending
      - initial J
      - how many parameters need to be estimated under H_0
  - Get used to them!

6 Appendix: The Chi-Squared Approximation for Pearson Tests

When we say that a statistic has approximately a χ_ν^2 distribution in large samples, for some number ν, what does that mean?

- The χ^2 is a *model* for the true sampling distribution of certain statistic
  - *It is wrong!*
  - Is it useful?
- Kind of like Central Limit Theorem for normal approximation to distribution of sample mean
  - If sample size is small, then approximation may be poor
  - As sample size grows larger, approximation becomes better on average

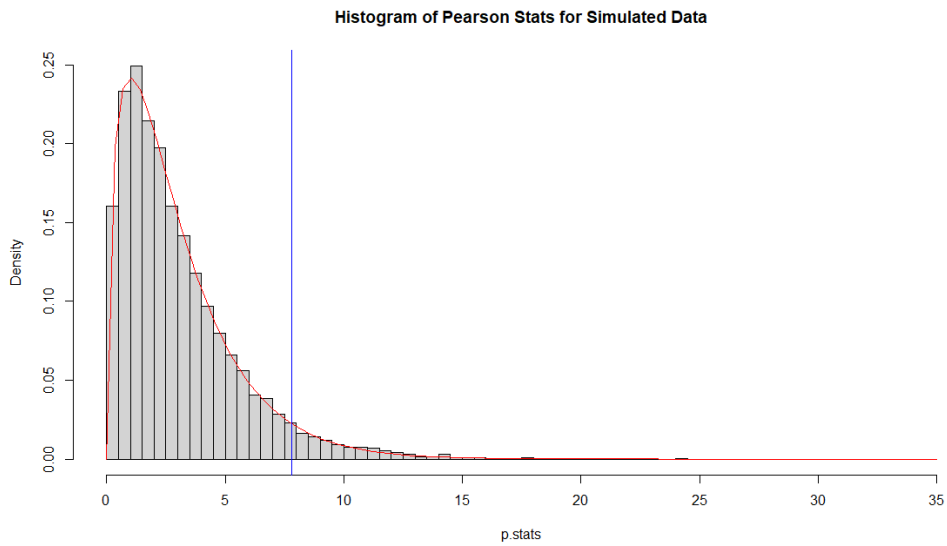
  - Even in very large samples (millions), still approximation, but *very, very good* (Difference between approximation and truth seen only in far-off decimal places.)

- Probabilities computed from approximate sampling distribution do not perfectly reflect behaviour of statistic upon repeated sampling and computation.
  - p-values computed from approximate sampling distribution are wrong
    - Reject H_0 when it is true with some probability $\neq \alpha$
  - Confidence levels are wrong: cover parameter with different probability than you think
  - But are they close enough that you can proceed with inferences and be assured that your answers are "good enough"?

- We can study the properties of certain tests, including test statistics, using simulation:
  1. Start from a model with parameters in it
  2. Have a null hypotheses that we want to test, which specifies values for those parameters
  3. **Use a computer to simulate a data set from this model using these special parameter values**
     (a) The null hypothesis is true for this data set!
  4. Compute a test statistic from the data in this data set
  5. **Repeat these two steps many times and store the results**
  6. Now we can use these many generated test statistic values to directly estimate the *actual* sampling distribution of the test statistic.
     (a) Does the model approximation fit the histogram of observed test statistic values?
     (b) Do certain critical values on model approximation cut off roughly the same probability on the histogram estimate?

- The program for this lecture includes code that simulates data from a multinomial with given probabilities and a given number of trials n.
  - Mimicking our example, our null hypothesis is that the probabilities follow a 9:3:3:1 ratio.
  - We generate 10,000 data sets (collections of four multinomial counts) using probabilities 9/16, 3/16, 3/16, 1/16.
  - We compute the Pearson statistic each data set (set of 4 counts) to compare its observed counts to the expected counts according to the null hypothesis.
  - These 10,000 computed statistics represent a random sample from the true sampling distribution of the Pearson statistic under the null hypothesis in this problem.

Figure 1: Histogram of simulated Pearson Statistics on data from Multinomial(1611, (9/16, 3/16, 3/16, 1/16)), with approximate χ^2_3 overlaid in red. The blue vertical line is the 95th percentile of the χ^2_3.



- Asymptotic theory suggests that these statistics should approximately follow a χ^2_3 distribution.
- So we make a histogram of the 10,000 Pearson statistics and overlay a χ^2_3 onto it to compare the fit.
- We also find that the proportion of these Pearson statistics that randomly fall beyond $\chi^2_{3,1-\alpha}$
  - This proportion should be very close to α if the model is a good enough approximation
  - In fact, we can treat each of these comparisons of a test statistic to $\chi^2_{3,1-\alpha}$ as a Bernoulli trial
    - Success = Pearson stat exceeds critical value
    - Hypothesize that the probability of success is α.
  - So we can perform a test of this hypothesis and make a confidence interval for the true probability.

- I ran the program using the original sample size, 1611 plants. See Figure 1 for the histogram and approximate distribution overlaid.

- Some of the simulated data are shown below as `rcount`. Each single data set is one column of this matrix. The `p.stats` object contains the Pearson goodness-of-fit test statistics applied to these data sets. The summary shows some relevant statistics for these 10000 samples from the true distribution of the statistic. Importantly, the mean is 3.02, whereas the mean of χ^2_ν is ν, the DF, which is 3 here. Pretty good match.

```
> rcount <- rmultinom(n=10000, size=samp.size, prob=c(9,3,3,1)/16)
> # Results are 10000 columns of 4 counts sampled from 9:3:3:1 ratio
> rcount[,1:6]
     [,1] [,2] [,3] [,4] [,5] [,6]
[1,]  866  914  896  918  878  888
[2,]  320  290  326  289  319  280
[3,]  301  306  300  301  303  342
[4,]  124  101   89  103  111  101
> # Run the Pearson stat on each column (MARGIN=2) of the data
> p.stats <- apply(X=rcount, MARGIN=2, FUN=pearson)
> summary(p.stats)
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
 0.007656  1.203876  2.350438  3.021296  4.132630 24.437478
> # Find fraction of simulations with Pearson values that exceed critical
> #   value.
> w <- sum(p.stats>qchisq(0.95, df=3))
> n <- 10000
> w/n
[1] 0.0523
> # Is this close to 0.05? Do the test!!!
> chisq.test(x=c(w,n-w), p=c(0.05,0.95), correct=FALSE)

        Chi-squared test for given probabilities

data:  c(w, n - w)
X-squared = 1.1137, df = 1, p-value = 0.2913

>
> # Wilson Score CI for the true probability
> library(package = binom)
> binom.confint(x = w, n = n, conf.level = 0.95, methods = "wilson")
  method   x     n   mean       lower      upper
1 wilson 523 10000 0.0523 0.04810587 0.05683796
```

The program calculates the proportion of these 10000 data sets that produced Pearson statistics that were large enough to (incorrectly) reject the null hypothesis: 0.0523. This "seems" pretty close to 0.05, the proportion we'd expect if the χ^2_3 were actually the right distribution. We can test whether this proportion might be estimating a true probability of 0.05. The Score test gives a test statistic $Z^2 = 1.11$ and a p-value 0.29, so we do not reject the null hypothesis that the test has exactly 0.05 type 1 probability. A 95% confidence interval for this probability is (0.048, 0.057).

- In conclusion with $n = 1611$, the Pearson test seems to have a sampling distribution under THIS null hypothesis that is pretty well approximated by χ^2_3
  - We don't know about other sample sizes, particularly *much* smaller ones

* We don't know about other probability configurations, particularly ones in which there could be one or more very rare categories.

7 Exercises

Instead of any analysis, read the appendix and play with the simulation program in the **Lecture 14 scripts.R**.

1. Decrease the sample size to something ridiculously tiny, $n = 9$.

   (a) What are the expected counts in this case? Are they at all "large" by the rules given in the lecture?

   (b) Simulate 5 data sets from the Multinomial(9, (9/16, 3/16, 3/16, 1/16)) distribution. Print out the results and just observe the variability in counts from one set to the other.

   (c) Run the full code to produce a plot and confidence interval for the probability of rejecting H_0 when it is true. Interpret the plot and confidence interval. Does χ^2_3 seem like a good fit for the actual sampling distribution of the Pearson statistic in this case?

2. Increase the sample size to 18, 27, 36, and so forth until you feel like the χ^2_3 starts to look like a pretty decent fit. Then increase it a few more times to be sure it wasn't just a fluke. Compute the expected counts and print out the plot and the confidence intervals for the rejection probability in each case. How large does the sample need to be, in your judgment, before you would be comfortable basing inferences on the large-sample approximation?

3. Now change the probabilities to 0.25, 0.25, 0.25, 0.25 and repeat the process in Exercise 2, starting from 9. Comment on the results and notice whether the approximation seems to look good at a larger or smaller sample size than before.

4. Now change the probabilities to 0.001, 0.010, 0.100, 0.889 and repeat the process in Exercise 2, starting from 9. Comment on the results and notice whether the approximation seems to look good at a larger or smaller sample size than before.

STATISTICS 475: Applied Discrete Data Analysis

Lecture 15: Contingency Tables

(B&L Section 3.2)

1 Problem to be solved

- We know the multinomial distribution for a single multicategory variable
- How do we model problems where two multicategory variables are measured on the same units?
  - We have done this before with two binary variables, a group variable and a response variable

2 $I \times J$ contingency tables

- Start with n units
- We have two categorical variables, X and Y, that are measured on each unit
  - X has I levels indexed by $i = 1, \ldots, I$
  - Y has J levels indexed by $j = 1, \ldots, J$
  - Observe a combination $(X = i, Y = j)$, which I will represent by "(i, j)"
    - Just like our 2×2 problem with $(0, 0), (0, 1), (1, 0), (1, 1)$.

<u>Example</u>: Energy Policy Opinions

From Ott and Longnecker (2001)[1].

> Random Samples of 200 individuals from major oil-producing and natural gas-producing states, 200 from coal-producing states, and 400 from other states participate in a poll of attitudes toward five possible energy policies. Each respondent indicated the preferred alternative from among the following:

[1] Ott, R.R. and Longnecker, M. (2001) *An Introduction to Statistical Methods and Data Analysis, 5th ed.* Pacific Grove CA, Duxbury, p.506

1. [`conserve`] Primarily emphasize conservation
2. [`oilgas`] Primarily emphasize domestic oil and gas exploration
3. [`solar`] Primarily emphasize investment in solar-related energy
4. [`nuclear`] Primarily emphasize nuclear energy development and safety
5. [`coal`] Primarily reduce environmental restrictions and emphasize coal-burning activities

In this context, we can take

- X = state, with levels `OILGAS`, `COAL`, and `OTHER`
- Y = policy, with levels listed above (for clarity, I have used small letters for policies and capitals for states)

There are 800 respondents, and each provides a measure of both X and Y.

- We usually summarize responses into counts for each *combination*
  - RV N_{ij} represents the population of possible counts of combination (i, j)
  - Observed count is n_{ij}
- Often represented as an $I \times J$ CONTINGENCY TABLE:

		Y				
		1	2	$\cdots$	J	Total
X	1	n_{11}	n_{12}	$\cdots$	n_{1J}	n_{1+}
	2	n_{21}	n_{22}	$\cdots$	n_{2J}	n_{2+}
	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	I	n_{I1}	n_{I2}	$\cdots$	n_{IJ}	n_{I+}
	Total	n_{+1}	n_{+2}	$\cdots$	n_{+J}	n

- MARGINAL COUNTS are the totals in a row or column
  - $n_{i+} = n_{i1} + n_{i2} + \ldots, n_{iJ}$, for each $i = 1, \ldots, I$ are the ROW MARGINAL COUNTS
  - $n_{+j} = n_{1j} + n_{2j} + \ldots, n_{Ij}$, for each $j = 1, \ldots, J$ are the COLUMN MARGINAL COUNTS
- Just extending the notation as we saw in binary (matched pairs) case, where $I = 2$ and $J = 2$

<u>Example</u>: Energy Policy Opinions (Lecture 15 scripts.R)

The counts from the energy policy opinion survey can be presented in a table as follows:

		policy					
		conserve	oilgas	solar	nuclear	coal	Total
state	OILGAS	n_{11} =50	88	56	4	2	n_{1+} =200
	COAL	59	20	52	3	66	200
	OTHER	161	40	188	5	6	400
	Total	n_{+1} =270	148	296	12	74	n =800

There are some interesting features in these counts...

2.1 Distributional Models for $I \times J$ table

Read Sections 3.2.1 and 3.2.2. They show different models based on multinomial distributions that can be used to represent the IJ counts.

> TLDR: Although two different models can be used to represent the counts, the end result is that *all parameter estimates, tests, and inferences are exactly the same under each model.* In other words, it doesn't matter at all which of the two models we use to represent the population from which the data came. Therefore, **I merely introduce the models here because we use each one's notation at different times.**

Single multinomial model This model focuses on modeling counts using JOINT PROBABILITIES of the outcome combinations of the two variables.

1. Gather n units and measure X and Y on each unit

   - A single multinomial distribution applies to all IJ counts
   - Joint probabilities for (X, Y) combinations are $\pi_{ij} = P(X = i, Y = j)$
     - $\sum_i \sum_j \pi_{ij} = 1$

		policy					
		conserve	oilgas	solar	nuclear	coal	Total
state	OILGAS	π_{11}	π_{12}	π_{13}	π_{14}	π_{15}	π_{1+}
	COAL	π_{21}	π_{22}	π_{23}	π_{24}	π_{25}	π_{2+}
	OTHER	π_{31}	π_{32}	π_{33}	π_{34}	π_{35}	π_{3+}
	Total	π_{+1}	π_{+2}	π_{+3}	π_{+4}	π_{+5}	1

   - PMF of the RVs $N_{11}, \ldots, N_{IJ}$ (also likelihood for $\pi_{11}, \ldots, \pi_{IJ}$) is

$$P(N_{11} = n_{11}, \ldots, N_{IJ} = n_{IJ}) \quad = \quad \frac{n!}{n_{11}! n_{12}! \ldots n_{IJ}!} \pi_{11}^{n_{11}} \pi_{12}^{n_{12}} \ldots \pi_{IJ}^{n_{IJ}}$$

   subject to the condition that $\sum_i \sum_j n_{ij} = n$.

- "Obvious" extension of the multinomial distribution to joint counts on two variables
- $\hat{\pi}_{ij} = n_{ij}/n$, the cell proportions relative to the total counts
  - Add to 1 across whole table:$\sum_i \sum_j \hat{\pi}_{ij} = 1$
- This is also the likelihood for this model

- Note that both sets of marginal counts are random
- This is very often (usually?) how data are collected, but it can vary by discipline

Product multimonial model This model focuses on modeling counts using the CONDITIONAL PROBABILITIES for outcomes of one of the variables, fixing one of the levels of the other variable

1. Gather fixed number of units in each row of the table and observe the other variable as a response

   - e.g., deliberately sample fixed numbers from groups, represented by levels of X
     - $n_{1+}, n_{2+}, \ldots, n_{I+}$ are known in advance
   - Experiments and surveys are often of this type
     - This is what was done in the energy policy example (a survey)
     - Also used in the "Fiber Enriched Crackers" example in the book (an experiment)

2. Each row has a separate multinomial distribution

   - Probabilities relate to categories of Y *given* the row of X
     - "Conditional probabilities", conditional on the row ("*given* the row category")
     - $P(Y = j|X = i) = \pi_{j|i}$ (Read the "|" as "given")
       - $\sum_{j=1}^{J} \pi_{j|i} = 1$ for each i
       - Conditional probabilities don't have to be the same in each row

		policy					
		conserve	oilgas	solar	nuclear	coal	Total
state	OILGAS	$\pi_{1\|1}$	$\pi_{2\|1}$	$\pi_{3\|1}$	$\pi_{4\|1}$	$\pi_{5\|1}$	1
	COAL	$\pi_{1\|2}$	$\pi_{2\|2}$	$\pi_{3\|2}$	$\pi_{4\|2}$	$\pi_{5\|2}$	1
	OTHER	$\pi_{1\|3}$	$\pi_{2\|3}$	$\pi_{3\|3}$	$\pi_{4\|3}$	$\pi_{5\|3}$	1
	Total	π_{+1}	π_{+2}	π_{+3}	π_{+4}	π_{+5}	1

   - Joint PMF within one row is standard multinomial. For row i:

$$P(N_{i1} = n_{i1}, N_{i2} = n_{i2}, \ldots, N_{iJ} = n_{iJ}|N_{i+} = n_{i+}) = \frac{n_{i+}!}{n_{i1}!n_{i2}!\ldots n_{iJ}!}\pi_{1|i}^{n_{i1}}\pi_{2|i}^{n_{i2}}\ldots\pi_{J|i}^{n_{iJ}}$$

   subject to the condition that $\sum_j n_{ij} = n_{i+}$

3. Since the RVs in one row are measured in different units from those in another row, the counts in different rows are independent of one another, so the joint PMF of *all* counts is just the product of the row PMFs:

$$\prod_{i=1}^{I} \frac{n_{i+}!}{n_{i1}!n_{i2}!\ldots n_{iJ}!} \pi_{1|i}^{n_{i1}} \pi_{2|i}^{n_{i2}} \ldots \pi_{J|i}^{n_{iJ}}$$

   - This is also the likelihood for this model
   - This is called a PRODUCT MULTINOMIAL MODEL
   - The MLE of $\pi_{j|i}$ is $\hat{\pi}_{j|i} = n_{ij}/n_{i+}$, the proportion of row-$i$ responses that are in column j

2.2 Independence of X and Y

- Two variables X and Y are INDEPENDENT if the outcome of one variable does not affect the probability distribution of outcomes for the other variable
  - For categorical variables, this means that probability distribution across the Y categories is the same regardless of which X category you check
    - In the context of the product multinomial model, each row has the same set of conditional probabilities.
    - i.e., for each j,

$$P(Y = j|X = 1) = P(Y = j|X = 2) = \ldots = P(Y = j|X = I) = P(Y = j)$$

    - Knowing X doesn't tell you anything about the probabilities of Y
  - Equivalently, just using notation, $\pi_{j|1} = \pi_{j|2} = \ldots = \pi_{j|I} = \pi_{+j}$
  - Can reverse Y and X in this definition and work with conditional distributions of rows, given columns
- In single multinomial, $\pi_{j|i} = \pi_{+j}$ implies that $\pi_{ij} = \pi_{i+}\pi_{+j}$
  - Same things happens in product multinomial if you define $\pi_{i+} = n_{i+}/n$

3 Testing for independence in a contingency table

- We often want to test for INDEPENDENCE between two categorical variables
- Conceptually, independence is a *lack of interaction* between two variables
  - *The conditional probability distribution across categories of one variable does not depend on the level of the other variable.*

- Mathematically, independence is a *simplification* of the probability structure in a multinomial model
  - Ordinarily, under the single multinomial model, we need $IJ - 1$ parameters to describe a contingency table, π_{ij}, $i = 1, \ldots, I$, $j = 1, \ldots, J$
    - The -1 is because they have to add to 1.
  - Under independence, each cell probability can be represented as a combination of marginal parameters, $\pi_{ij} = \pi_{i+}\pi_{+j}$
    - Don't need each cell parameter, just marginal probabilities
    - There are I parameters π_{i+}, but they must add to 1, so $I - 1$ parameters are needed to represent rows
    - There are J parameters π_{+j}, but they must add to 1, so $J - 1$ parameters are needed to represent columns
    - So $I + J - 2$ parameters are needed to describe the probabilities in the entire table
- Thus, independence—the simpler model—is always the null hypothesis in a test based on comparing model fits.
- $H_0 : X$ and Y are independent vs. $H_a : X$ and Y are not independent
  - Formally,

$$H_0 : \pi_{ij} = \pi_{i+}\pi_{+j} \text{ for } \textbf{all } i, j \text{ combinations}$$
$$H_a : \pi_{ij} \neq \pi_{i+}\pi_{+j} \text{ for } \textbf{at least one } i, j \text{ combination}$$

- There are two basic philosophies for testing for independence:
  1. Compare observed counts to what would be expected under the null hypothesis using a Pearson statistic
  2. Compare a model fit under independence with a model fit without it using a LR test
- If we reject H_0, then we conclude that the two variables are *not independent*, or that they *are* ASSOCIATED
  - "Association" is often taken to be the opposite of "independence"

3.1 Pearson Test

- Regardless of the model, the expected (mean) count cell (i, j) of the table is $\mu_{ij} = n\pi_{ij}$ for $i = 1, \ldots, I;\ j = 1, \ldots, J$ if we impose no restrictions on the probabilities.
- *Under the null hypothesis of independence*, since $\pi_{ij} = \pi_{i+}\pi_{+j}$, the expected cell count in cell (i, j) can be written as a function of marginal parameters: $\mu^0_{ij} = n\pi_{ij} = n\pi_{i+}\pi_{+j}$

- MLE of this is

$$\hat{\mu}_{ij}^0 = n * \frac{n_{i+}}{n} * \frac{n_{+j}}{n} = \frac{n_{i+}n_{+j}}{n}$$

- Construct Pearson statistic comparing observed counts to those expected under H_0:

$$X^2 = \sum_{i=1}^{I}\sum_{j=1}^{J} \frac{(n_{ij} - \hat{\mu}_{ij}^0)^2}{\hat{\mu}_{ij}^0}$$

- Reject H_0 if $X^2 > \chi^2_{???}$.
  - How many DF? Why?

3.2 Likelihood ratio test

We describe this test under the single multinomial model, but the results are the same under product multinomial

- Model under H_a is just the single multinomial with unrestricted π_{ij}s
  - Find MLEs and evaluate the likelihood at the MLEs
- Model under H_0 substitutes $\pi_{i+}\pi_{+j}$ for each π_{ij}
  - Find MLEs and evaluate the likelihood at the MLEs
- Compute $-2\log(\Lambda)$
- Reject H_0 if $X^2 > \chi^2_{???}$
  - How many DF? Why?

<u>Example</u>: Energy Policy Opinions (Lecture 15 scripts.R, EnergyPolicy.csv)

We will test whether policy preference is independent of kind of state the respondent resides in. As a reminder, here is the table:

		policy					
		conserve	oilgas	solar	nuclear	coal	Total
state	OILGAS	50	88	56	4	2	200
	COAL	59	20	52	3	66	200
	OTHER	161	40	188	5	6	400
	Total	270	148	296	12	74	800

The data are in the spreadsheet, **EnergyPolicy.csv**. First we read in the data. R rearranges the levels alphabetically, so I undo that first so that I can reproduce the same table as above. This isn't necessary for statistical reasons; it's just for looks. The `xtabs()` function then creates a table that can be printed and analyzed further.

```
> set1 <- read.csv(file="C:\\data\\EnergyPolicy.csv")
> set1
    state   policy count
1  OILGAS conserve    50
2  OILGAS   oilgas    88
3  OILGAS    solar    56
4  OILGAS  nuclear     4
5  OILGAS     coal     2
6    COAL conserve    59
7    COAL   oilgas    20
8    COAL    solar    52
9    COAL  nuclear     3
10   COAL     coal    66
11  OTHER conserve   161
12  OTHER   oilgas    40
13  OTHER    solar   188
14  OTHER  nuclear     5
15  OTHER     coal     6
> # Note order of levels in the two factors
> levels(set1$state)
[1] "COAL"   "OILGAS" "OTHER"
> levels(set1$policy)
[1] "coal"     "conserve" "nuclear"  "oilgas"   "solar"
>
> # If I want to match order from table in class,
> #   need to rearrange levels using
> #   factor(<original factor object>,
> #     levels=levels(<original factor object>)[<new ordering>])
>
> set1$state <- factor(set1$state,
+                      levels=levels(set1$state)[c(2,1,3)])
> set1$policy <- factor(set1$policy,
+                       levels=levels(set1$policy)[c(2,4,5,3,1)])
> levels(set1$state)
[1] "OILGAS" "COAL"   "OTHER"
> levels(set1$policy)
[1] "conserve" "oilgas"   "solar"    "nuclear"  "coal"
>
> op.table <- xtabs(formula = count ~ state + policy, data = set1)
> op.table
        policy
state    conserve oilgas solar nuclear coal
  OILGAS       50     88    56       4    2
  COAL         59     20    52       3   66
  OTHER       161     40   188       5    6
```

There are several functions that can compute the Pearson test and fewer that can compute the LR test. I show a few of them below:

```
> # summary() on "table" or "xtabs" objects does Pearson test
> class(op.table)
[1] "xtabs" "table"
> summary(op.table)
Call: xtabs(formula = count ~ state + policy, data = set1)
Number of cases in table: 800
Number of factors: 2
Test for independence of all factors:
        Chisq = 289.22, df = 8, p-value = 8.082e-58
        Chi-squared approximation may be incorrect
>
>
> # Both Pearson and LR tests
> library(package = vcd)
> assocstats(x = op.table)
                    X^2 df P(> X^2)
Likelihood Ratio 249.17  8        0
Pearson          289.22  8        0

Phi-Coefficient   : NA
Contingency Coeff.: 0.515
Cramer's V        : 0.425
>
>
> # Pearson Test
> ind.test <- chisq.test(x=op.table, correct=FALSE)
Warning message:
In chisq.test(x = op.table, correct = FALSE) :
  Chi-squared approximation may be incorrect
> ind.test

        Pearson's Chi-squared test

data:  op.table
X-squared = 289.22, df = 8, p-value < 2.2e-16
```

The Pearson statistic is $X^2 = 289.22$ which has a *tiny* p-value on χ^2_8. We therefore reject the null hypothesis of independence and conclude that people's opinions about energy policy are associated with the kind of energy their state produces. The LR test ($-2\log(\Lambda) = 249.17$) leads to the same conclusion.

---

4 Follow-up to the test

- Rejecting the null hypothesis of independence doesn't tell you much
  - The conditional distribution across the Y levels is not the same in every X level.
    - We know nothing about *how they differ*
  - *Want to figure out which rows have different conditional probabilities for which columns*
- Can do a few things to understand the result better
  - Visually compare estimated conditional probabilities, $\hat{\pi}_{j|i} = n_{ij}/n_{i+}$ across rows
    - Intuitive but not objective and no quantification of uncertainty
  - Can look at certain odds ratios (complicated unless there are specific questions)
    - Subtables: Select any two rows and columns for OR
      - Reduces total sample size by excluding many counts
    - Partitions: Make two groups of rows and 2 groups of columns, partitions table into 4 cells
      - Retains full sample size, but ignores possible differences among categories within a partition
    - Complicated and potentially arbitrary unless there are specific questions that subtables or partitions can answer
      - **Do not choose subtables or partitions because they "look like" they have big associations!**
  - Look at deviations between observed and expected values...

4.1 Standardized Pearson Residuals

- In linear regression, RESIDUALS measure how far an observed response lies from the mean estimated by the model.
- We can do the same thing, comparing how far our observed counts lie from our expected count under the independence model
  - The deviations between the observed and expected counts under independence is $n_{ij} - \hat{\mu}^0_{ij}$
- Large differences should mean that the cell deviates from independence
  - However, larger counts have larger variances
  - A difference of, say, 5 is more of a deviation if the expected count is 1 than when it is 100

- So we divide residuals by their estimated standard errors to standardize them all to have variance 1:

$$r_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}^0}{\sqrt{(1 - n_{i+}/n)(1 - n_{+j}/n)\hat{\mu}_{ij}^0}}$$

  - This is called a STANDARDIZED PEARSON RESIDUAL

- Kind of like creating a fancy version of a Z-score
  - The distribution of a standardized Pearson residual can be *approximated* by the standard normal
    - Values beyond, say, ± 2 are <u>somewhat unusual</u> if independence really holds
      - Should occur only in about 5% of cells
    - Values beyond, say, ± 3 are <u>exceedingly rare</u> when independence holds
      - Should occur in only about 0.3% of cells
  - So look for "large" magnitudes in the table.
    - Focus on the largest magnitudes values first

<u>Example</u>: Energy Policy Opinions (Lecture 15 scripts.R, EnergyPolicy.csv)

Below are the estimated conditional probabilities for each type of state, the expected counts in each cell, and the standardized Pearson residuals. The `prop.table()` function computes estimates of conditional probabilities, conditioning on each row (`margin=1`), or column (`margin=2`). The `chisq.test()` function that provided the Pearson test also produces expected counts and standardized residuals as different components in the stored object.

```
> # Look at estimated conditional probabilities in each row
> prop.table(op.table, margin=1)
        policy
state    conserve oilgas  solar nuclear   coal
  OILGAS   0.2500 0.4400 0.2800  0.0200 0.0100
  COAL     0.2950 0.1000 0.2600  0.0150 0.3300
  OTHER    0.4025 0.1000 0.4700  0.0125 0.0150
> # Expected counts
> ind.test$expected
        policy
state    conserve oilgas solar nuclear coal
  OILGAS     67.5     37    74       3 18.5
  COAL       67.5     37    74       3 18.5
  OTHER     135.0     74   148       6 37.0
> # Standardized Pearson residuals
> round(ind.test$stdres, digits=1)
        policy
state    conserve oilgas solar nuclear coal
```

```
OILGAS      -3.0   10.7  -3.0     0.7 -4.6
COAL        -1.5   -3.6  -3.7     0.0 13.4
OTHER        3.9   -6.2   5.9    -0.6 -7.6
```

The conditional probabilities are easy to interpret:

- People in states that produce a particular energy resource tend to prefer policy that favours those resources more than other people do
- People who live in states that don't produce coal, oil, or gas tend to prefer conservation or solar power more than other people do.

The standardized residuals tell the same story, although in a somewhat more complicated way

- There are huge positive residuals for coal preference in coal states and oil/gas preference in oil/gas states.
  - Positive residual means higher observed counts than expected under the model of independence.
- These are naturally going to cause smaller negative residual as a counterbalance because above-expected counts somewhere must produce below-expected counts elsewhere.
- The preference for conservation and solar power in other states is apparent from the positive residuals

---

5 Notes

- Pearson test is a score test, so it, and the LR, are both "good" tests (better than Wald).
- As always, the χ^2 distributions are accurate in large samples, but may not be in small samples.
  - A typical recommendation is that *all* expected cell counts must be at least 5: $\hat{\mu}_{ij}^0 \geq 5$ for all i, j
  - This is usually fine, but the approximation will still hold reasonably well as long as most expected cell counts are at least 5 and just a few are still at least 1
  - See Section 6.2 of B&L for methods that work well regardless of sample size, but are a bit more cumbersome to use.

- The standardized Pearson residuals are also somewhat sensitive to small expected counts.
  - Residuals with large magnitudes can be created when the expected count is close to 0 and the observed count is anything other than 0.
  - Don't put too much faith in them unless expected counts are at least, say, 2 or more (a guess)
- Sections 3.3 and 3.4 of the book tell you how to fit regression models to multinomial counts, including those in contingency tables.
  - Allows possible models to express forms of association other than independence
  - Allows us to consider more than two categorical variables
  - These models are cumbersome to work with, so we will skip them (I rarely use them)
  - Chapter 4 will show us how to fit more interpretable models that achieve mostly the same goals

6 What to learn from this

- Contingency tables are *very* popular ways to express counts of responses with respect to two categorical variables
- Independence/association tests are *extremely* popular.
  - Pearson test is most common.
  - Understand why expected cell counts are computed as $n\pi_{i+}\pi_{+j}$
- Limited things one can do as follow-up
  - Estimated conditional distributions for each row
  - Standardized Pearson residuals

7 Exercises (due when announced)

The data below are from a study of birth defects among children (yes or no), cross-classified by the mother's average daily alcohol consumption as of the third month of pregnancy (average drinks per day):

		Birth Defects		
		Yes	No	Total
Drinks	0	48	17,066	17,114
	< 1	38	14,464	14,502
	1–2	5	788	793
	3–5	1	126	127
	≥ 6	1	37	38
	Total	93	32,481	32,574

1. Use `R` to compute the proportion of children with birth defects in each group. Comment on whether there seems to be any association between alcohol consumption and the probability of birth defects.

2. A conspiracy theorist would write a meme showing only the "Yes" column of numbers. They would argue that health care providers should stop telling women to avoid alcohol when pregnant, because there were actually *fewer* children with birth defects as the mothers drank more and more. Knowing the full table, explain why this is conclusion is wrong.

3. Analyze these data:

   (a) Perform the Pearson chi-square test for independence using these data. Write out all parts of the test (hypotheses, test statistic, p-value, decision rule, decision, and conclusion).

   (b) Notice that this tests has 4 DF. Explain this number in terms of the number of conditional probabilities of birth defects, given drinks, that must be estimated under the null and alternative hypotheses.

   (c) Follow this with standardized residuals and interpret the results.

4. Repeat the computation of the test statistic and p-value using LR methods. Report these results.

5. Notice that this is a model with a binary response, so we could have used logistic regression.

   (a) Fit the logistic regression treating the number of drinks as a factor. You may need to rearrange the data to do this. Report the model.

   (b) Compute the test statistic and p-value for the LR test for whether there is any effect of the drink category on the probability of birth defects. Be sure that these categories form a nominal factor. Write out all parts of the test. Specifically, write out the hypotheses in terms of logit models.

(c) If you have done this right, you should find that the two LR tests are identical. Explain why the two null hypotheses tested in parts (3) and (4) are the same.

6. Notice that the "Drinks" variable is ordinal. Approximate the number of drinks in each group, from top to bottom, using numbers 0, 0.5, 1.5, 4, and 7, respectively. Perform a new logistic regression, modeling $\pi = P(\text{Birth Defect})$ against the number of drinks.

   (a) Write out the estimated model

   (b) Explain what the "slope" measures in the model

   (c) Perform a LR test for whether the probability of birth defects changes as the number of drinks changes. Write out all parts of the test. Specifically, write out the hypotheses in terms of logit models.

   (d) If you have done this right, you should find that this test has a somewhat smaller p-value than the others. Speculate why that might be.

STATISTICS 475: Applied Discrete Data Analysis

Lecture 18: Intro to the Poisson Distribution

(B&L Section 4.1)

1 Problem to be solved

- Thus far, we have studied two models: Binomial and Multinomial
  - These are very similar distributions (Binomial is just Multinomial limited to $J = 2$ outcomes)
- Both are counts of category occurrences in n identical trials (n could be 1)
- *The key with these counts is that there must be some kind of trials, each of which ends in ONE particular outcome*
  - We are limited to one outcome category per trial
  - Every trial is observed, and it is either a success or a failure
  - Results in counts $\leq n$, and we know what n is.
- Not all counts are of this form
  - Number of cars crossing a bridge in an hour
  - Number of weeds in a plot of crop land
  - Number of moles on a person's body
- These counts are free to vary between 0 and **no <u>specific</u> fixed upper limit**
  - There may be *practical* limits, counts you cannot achieve
    - Can't have 100,000,000 cars crossing the bridge in an hour
    - *But can't specify what the maximum <u>is</u>.*
- Also, there is not a clearly defined notion of trial where failure is observed

- What is the opposite of a car crossing a bridge, a weed in a plot, or a mole on a body

- We need a new distributional model to handle counts of this type
  - There are several
  - Simplest and most popular is the POISSON DISTRIBUTION

2 The Poisson Distribution

- The key feature of unrestricted counts is that they can potentially take on *any* non-negative integer value
  - A response variable Y (the population of possible counts) can take values $0, 1, 2, \ldots$ with <u>no fixed, known upper limit</u>

- Probability distribution typically has higher probabilities within a limited range where counts are likely for the phenomenon, low probabilities elsewhere
  - Recall that Normal puts positive probability *everywhere*.
    - That means that Normal is the wrong model for pretty much anything, but it may be a *useful* model for many measurements with a mound-shaped distribution
  - Poisson similarly puts positive probability on *all* non-negative integers, no matter how large they are, but with a hump around most likely values
    - May be wrong, but it is a *model*. It doesn't have to be perfect to be useful.

- If Y is a Poisson RV, then the PMF of Y is

$$P(Y = y) = \frac{e^{-\mu}\mu^y}{y!}, \; y = 0, 1, 2, \ldots$$

  - Write $Y \sim Po(\mu)$
  - $\mu > 0$ is the only parameter

- Properties of $Po(\mu)$
  1. $E(Y) = Var(Y) = \mu$
     - Unlike normal where variance is unrelated to the mean
     - This is an important property of $Po(\mu)$, and its main limitation
  2. If $Y_1, Y_2, \ldots, Y_M$ are independent Poisson RVs, where $Y_m \sim Po(\mu_m)$ for $m = 1, \ldots, M$, then $\sum_{m=1}^{M} Y_m \sim Po(\sum_{m=1}^{M} \mu_m)$
     - Useful for modeling totals from several counts
     - e.g., monitor bridge for an hour, then another hour

        - If each count has a Poisson distribution, then the total does as well

3. As μ gets larger, the Poisson distribution looks more and more like a normal distribution with the same mean and variance
    - See Figure 1 for examples of Poisson distributions and corresponding normal distributions with the same mean and variance.

- Remember, the Poisson distribution is a *model*, and may not be a perfect representation for a given real set of data.
    - See the "Cars at an intersection" example in the book (also depicted in Figure 2)
    - Even when not a perfect fit, it may still be useful

3 Estimation and Inference

- Start with a sample of n counts, each from a Poisson distribution with *the same mean.*
    - $y_1, y_2, \ldots, y_n$ independently taken from $Po(\mu)$
        - Each response is a count
        - Could have $n = 1$!
    - For example
        - Count cars at the same intersection on many days
        - Count weeds in several plots of land that have been treated the same way
        - Count moles on random people ~~at the beach~~ at a dermatology clinic
- The Poisson likelihood for estimating the mean (and variance) parameter μ from a random sample of counts is just the product of PMFs evaluated at the observed counts, $y_1, \ldots, y_n$:

$$L(\mu; y_1 \ldots, y_n) = \prod_{i=1}^{n} \frac{e^{-\mu}\mu^{y_i}}{y_i!}$$

- MLE for μ turns out to be $\hat{\mu} = \bar{y}$
    - $Var(\hat{\mu}) = \mu/n$, so estimated standard error for mean is $\sqrt{\widehat{Var}(\hat{\mu})} = \sqrt{\hat{\mu}/n}$
    - Compare to normal case: $\hat{\mu} = \bar{y}$, but $\widehat{Var}(\hat{\mu}) = s^2/n$, where s^2 is the sample variance
        - Normal distribution has separate parameters for mean and variance; Poisson reuses μ
- Can use these facts to develop tests and confidence intervals for μ.

Figure 1: Examples of Poisson distribution PMFs for various values of μ (black bars), with corresponding normal distributions $N(\mu, \mu)$ overlaid (red curve)

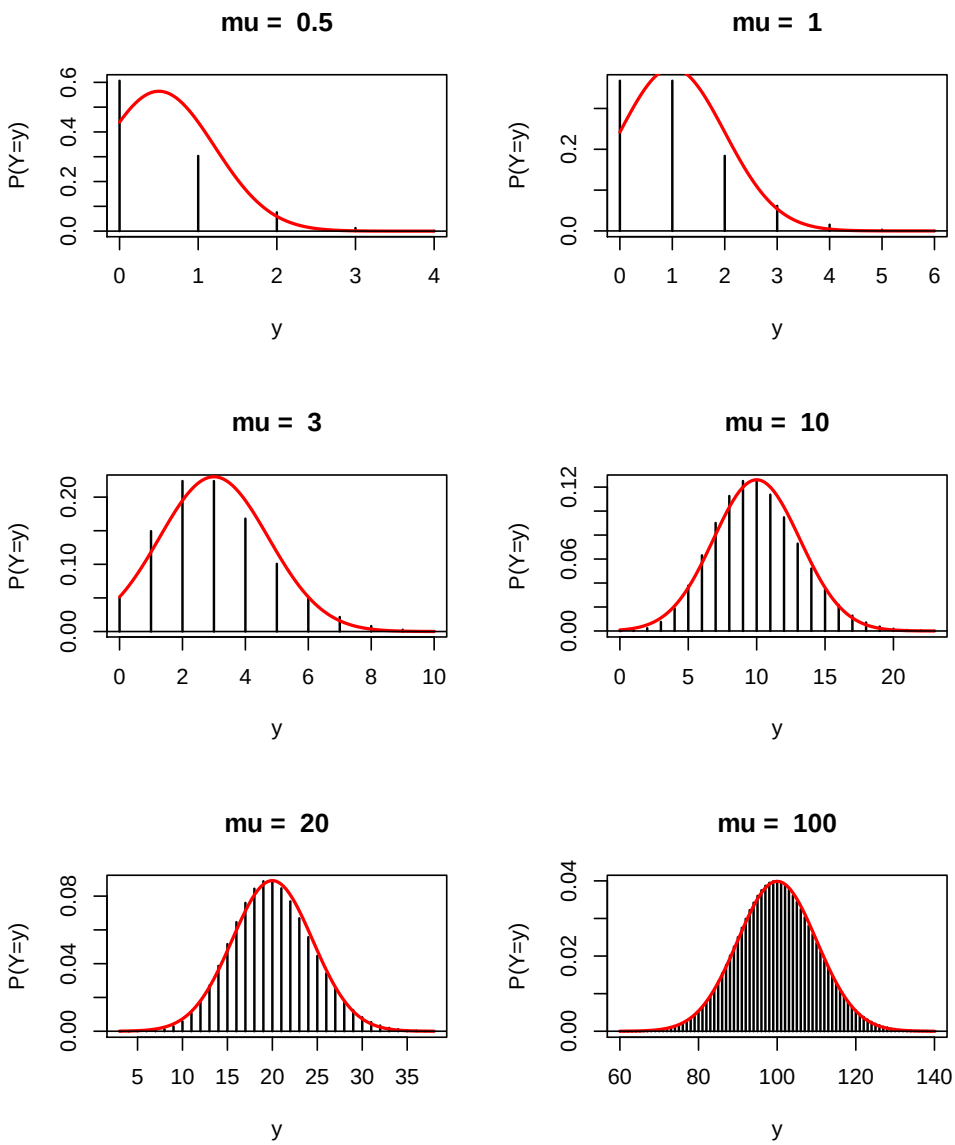
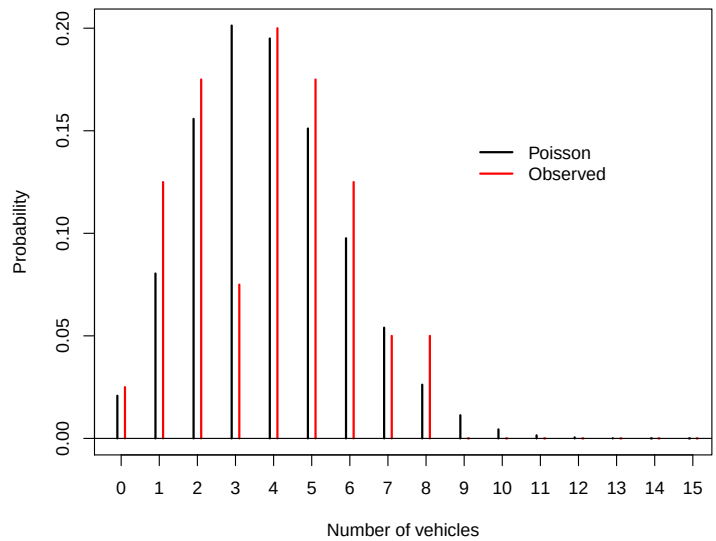


Figure 2: Histogram of observed counts (red) for **Cars at an Intersection** example with best-fitting Poisson distribution (black) overlaid. Taken from book.



Hypothesis tests for μ

- As in normal case, test $H_0 : \mu = \mu_0$
  - $H_a : \mu > \mu_0$
  - $H_a : \mu < \mu_0$
  - $H_a : \mu \neq \mu_0$
- **Wald test** (Estimate standard error at MLE):

$$Z_W = \frac{\hat{\mu} - \mu_0}{\sqrt{\hat{\mu}/n}}$$

- **Score test** (Estimate standard error at H_0):

$$Z_0 = \frac{\hat{\mu} - \mu_0}{\sqrt{\mu_0/n}}$$

- Rejection regions and p-values found from *standard normal distribution*
  - *not a t-test!*
  - t-distribution results specifically from having a sum of squares (e.g., s^2) in the denominator

- **LR Test** (compare log-likelihoods at MLE and at H_0):

$$-2\log(\Lambda) = 2n[(\hat{\mu} - \mu_0) + \hat{\mu}\log(\hat{\mu}/\mu_0)]$$

  - Compare to χ^2_1, because we estimate 1 parameter under H_a and none under H_0.

Confidence intervals for μ

- The $100(1-\alpha)\%$ Wald confidence interval is

$$\hat{\mu} \pm Z_{1-\alpha/2}\sqrt{\hat{\mu}/n}$$

- The score interval is derived from finding the values of c for which $H_0 : \mu = c$ is not rejected:

$$\left(\hat{\mu} + \frac{Z^2_{1-\alpha/2}}{2n}\right) \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{\mu} + Z^2_{1-\alpha/2}/4n}{n}}.$$

- Both of these need $\hat{\mu}$ large enough that normal approximation is reasonable
  - Also need large enough n unless μ is so large that normal is *very* good fit
  - Score is not bad even for $n = 5$ and $\mu \approx 3$ or more
- The LR interval is the set of values for c for which $-2\log[L(\mu_0|y_1\ldots,y_n)/L(\hat{\mu}|y_1\ldots,y_n)] \leq \chi^2_{1,1-\alpha}$.
  - No closed-form solution, not usually used.
- The exact interval is developed using logic similar to what was used for the Clopper-Pearson interval for the binomial probability:

$$\chi^2_{2n\hat{\mu},\alpha/2}/2n < \mu < \chi^2_{2(n\hat{\mu}+1),1-\alpha/2}/2n$$

  - Only interval that is guaranteed to have true confidence level at least $1-\alpha$, but may be excessively wide
- See Fig 4.2 in the book for some confidence level examples, along with comparisons to t-based intervals assuming normality

<u>Example</u>: Age of Abalone (Lecture 18 scripts.R, abalone.csv)

From the source where I got these data,http://archive.ics.uci.edu/dataset/1/abalone:[1]

[1]Interesting information on abalones can be found here: https://www.youtube.com/watch?v=WbEmnfY-niU . I recommend that you look at this video if you have never seen an abalone before. WARNING: It is a little gross.

> "Predicting the age of abalone from physical measurements. The age of abalone is determined by cutting the shell through the cone, staining it, and counting the number of rings through a microscope – a boring and time-consuming task. Other measurements, which are easier to obtain, are used to predict the age. Further information, such as weather patterns and location (hence food availability) may be required to solve the problem."

The variables in the data set are as follows, also from the same website:

- `Sex` / categorical / – / M, F, and I (infant) (These are coded in the data as 1, 2, and 0, respectively [*Coded as numeric! Need to change this in code.*])
- `Length` / numerical / mm / Longest shell measurement
- `Diameter` / numerical / mm / perpendicular to length
- `Height` / numerical / mm / with meat in shell
- `Whole` weight / numerical / grams / whole abalone
- `Shucked` weight / numerical / grams / weight of meat
- `Viscera` weight / numerical / grams / gut weight (after bleeding)
- `Shell` weight / numerical / grams / after being dried
- `Rings` / integer count / – / +1.5 gives the age in years

We will be looking at ways to predict $Y=$ the number of rings in an abalone based on other measurements. We first ask what model we should use for the counts. Notice that

1. the count of rings is not a result of a number of successes in a particular number of trials,
2. the presence of a ring does not have a measurable opposite "failure,"
3. and the number of rings has no specific number as its upper bound, although there are practical limits due to biology.

Thus, we cannot use a binomial model for these counts. A Poisson model is more appropriate.

First, we take a look at the number of rings as a measurement by itself and see whether it might be modeled with a Poisson distribution.

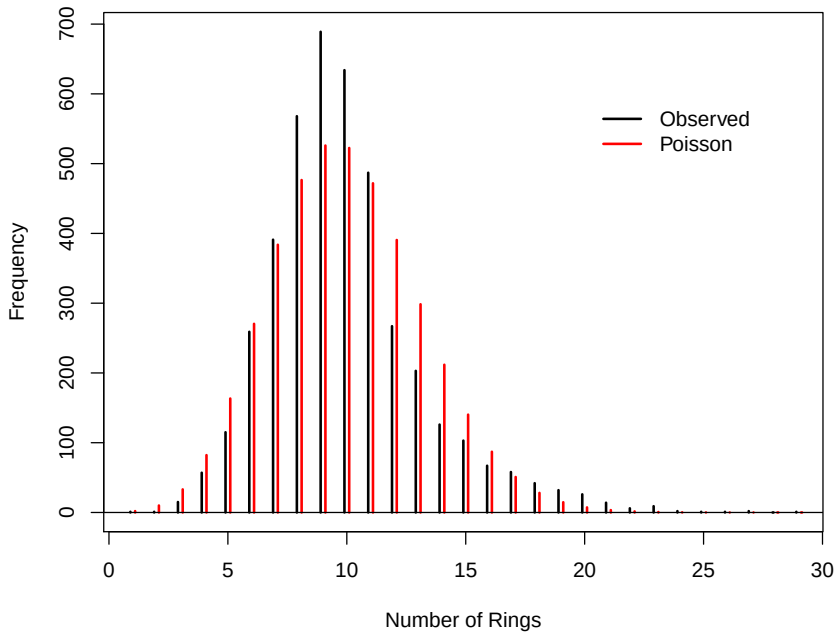
```
> aba <- read.csv(file  = "abalone.csv")
> head(aba)
  Sex Length Diameter Height  Whole Shucked Viscera Shell
1   1  0.455    0.365  0.095 0.5140  0.2245  0.1010 0.150
2   1  0.350    0.265  0.090 0.2255  0.0995  0.0485 0.070
3   2  0.530    0.420  0.135 0.6770  0.2565  0.1415 0.210
```

```
4   1  0.440    0.365  0.125 0.5160  0.2155  0.1140 0.155
5   0  0.330    0.255  0.080 0.2050  0.0895  0.0395 0.055
6   0  0.425    0.300  0.095 0.3515  0.1410  0.0775 0.120
  Rings
1    15
2     7
3     9
4    10
5     7
6     8
> summary(aba)
      Sex              Length          Diameter
Min.   :0.0000   Min.   :0.075   Min.   :0.0550
1st Qu.:0.0000   1st Qu.:0.450   1st Qu.:0.3500
Median :1.0000   Median :0.545   Median :0.4250
Mean   :0.9916   Mean   :0.524   Mean   :0.4079
3rd Qu.:2.0000   3rd Qu.:0.615   3rd Qu.:0.4800
Max.   :2.0000   Max.   :0.815   Max.   :0.6500
       Height            Whole            Shucked
Min.   :0.0000   Min.   :0.0020   Min.   :0.0010
1st Qu.:0.1150   1st Qu.:0.4415   1st Qu.:0.1860
Median :0.1400   Median :0.7995   Median :0.3360
Mean   :0.1395   Mean   :0.8287   Mean   :0.3594
3rd Qu.:0.1650   3rd Qu.:1.1530   3rd Qu.:0.5020
Max.   :1.1300   Max.   :2.8255   Max.   :1.4880
      Viscera            Shell             Rings
Min.   :0.0005   Min.   :0.0015   Min.   : 1.000
1st Qu.:0.0935   1st Qu.:0.1300   1st Qu.: 8.000
Median :0.1710   Median :0.2340   Median : 9.000
Mean   :0.1806   Mean   :0.2388   Mean   : 9.934
3rd Qu.:0.2530   3rd Qu.:0.3290   3rd Qu.:11.000
Max.   :0.7600   Max.   :1.0050   Max.   :29.000
> (n <- nrow(aba))
[1] 4177
> (mean.ring <- mean(aba$Rings))
[1] 9.933684
> (var.ring <- var(aba$Rings))
[1] 10.39527
```

We see that there are 4177 observations, and some variables have *unbelievably small minima!* Also, I'm pretty sure that the units of measurement on the website do not match what is in the file (abalone are *not* tiny creatures smaller than one cubic millimeter!). Nothing we can do about that, other than to be skeptical about the information on the website.

We consider fitting a Poisson distribution with mean μ to the number of rings. The MLE for μ is $\hat{\mu} = \bar{y} = 9.9$.

Figure 3: Frequency histogram of abalone Rings data, with expected counts from Poisson with the same mean.



Count data often have sample variances that are much bigger than their respective means. This is a sign of OVERDISPERSION, a common inadequacy of the Poisson model. See Section 5.3 of the book—we will get to this later. Here the variance is 10.4, which seems "close" to 9.9. We don't yet have the pieces in place to test whether this difference is "significant"—doing so requires learning yet another model for counts, the "negative binomial" also covered in Section 5.3 of the book.

A plot of the data (see code in the program) is shown in Figure 4. The extra variance in the observed data is hard to see at first, because the center peak is actually higher than the Poisson predicts. This would seem to suggest a *lower* variance for the actual data, since more observations are tighter to the middle. However, there are proportionately much higher numbers of large observations in the data than the Poisson predicts (even though the histogram bars are all low), and this has a high impact on the variance. The program also shows code for manually computing the confidence intervals. No matter which interval one computes, they lead to about the same answer, 9.84–10.03, because of the very large sample size.

We will do more with this data set in later lectures.

---

4 Notes

- Although I show them here, confidence intervals for mean counts assuming Poisson distributions are not often done.
  - People tend to like to use normal-based t-methods instead, even when data are "so discrete" that normality is a very poor model
    - This isn't necessarily good, by the way.
    - The reason people do it is that overdispersion is *so* common that intervals based on assuming that the Poisson model is **correct** are often way too short
      - They do not cover the parameter in 95% of samples.
      - Allowing the variance to be estimated separately from the mean improves coverage.
- However, a Poisson CI can be computed even when $n = 1$!
  - You can't do *that* with a normal distribution!

5 Conclusions: What to learn

1. We have a new distributional model. Learn it.
2. **Know the difference between a count that could be modeled by a binomial distribution and a count that could be represented by a Poisson.**
3. The mean and variance are equal *in the model*. This is often not true in the populations from which data are drawn.
4. Reinforce your understanding of Wald, LR, and score inferences.

Table 1: Salamander count data.

Years after burn	12	12	32	20	20	27	23	19	23	26
Salamanders	3	4	8	6	10	5	4	7	2	8
Years after burn	21	3	8	35	2	19	8	25	33	35
Salamanders	6	0	2	6	1	5	1	5	4	10

6 Exercises (due when announced)

1. For each of the following cases, indicate whether the count response would be better modeled with a Poisson or a binomial distribution

   (a) During a 2-hour study session, you count how many texts you receive.

   (b) During a 2-hour study session, you count how many of your 10 best friends message you.

   (c) Out of 20 days on campus, you count how many days have some connection issue with the campus wifi.

   (d) In a football game, you count the number of times a placekick is attempted

   (e) Over a month, you count how many times your best friend (or partner or roommate) loses their phone (or keys or wallet).

   (f) You randomly sample 40 students and record how many have a part-time job.

2. The data in the table are from a study of salamanders where counts were taken on plots of fixed size sampled from prairie regions in Manitoba, Canada.[2] The prairies are subject to periodic controlled burning to reduce the chance of uncontrolled wildfires, so different plots had been burned in different numbers of years past. Ignore this feature for now.

   (a) Plot the count of salamanders per plot along with the Poisson model using its MLE for μ as in Figure 3. Comment on the fit of the model with the same mean. Note that if you use code similar to that given in the Lecture 18 Scripts, you will need to add 1 to the variable in the `tabulate()` function, because there are some counts of 0 in the data, and you will need to amend the range over which expected counts are taken to more closely resemble the range of the actual data.

   (b) Assume that the Poisson model is reasonable. Report the MLE of the mean and find a 95% confidence interval for the mean number of salamanders per plot

   (c) Report the sample variance and find a 95% confidence interval for it using standard methods that assume normality of the counts. Does this confidence interval overlap with the CI for the mean? Does this offer any clues as to whether overdispersion may be a problem here? Explain.

[2] Data kindly provided by Dr. Carl Schwarz, Department of Statistics and Actuarial Science, Simon Fraser University.

STATISTICS 475: Applied Discrete Data Analysis

Lecture 19: Poisson Regression Models

(B&L Section 4.2.1, 4.2.2)

1 Problem to be solved

- We learned the binomial distribution for counts of successes from a fixed number of Bernoulli trials or for binary responses
  - Then we learned how to do regression to model the probability of success against explanatory variables using a linear model for the log-odds of success
- We learned the multinomial distribution for counts from a fixed number of multinomial trials or multicategory responses
  - Then we (could have) learned how to do regression to estimate the probability of *each category* against explanatory variables
  - This is the subject of the missing Lectures 16 and 17.
- We have just learned the Poisson distribution for counts that are *not* the result of observing categories from a fixed number of trials
  - Now we need to develop regression models for these counts
  - It turns out that these models can very adequately substitute as models for counts from binomial or multinomial models!
    - We get to learn a way to deal with multinomial regression after all... :)

2 The Poisson regression model

- The basic Poisson distribution model assumes that the true mean count is the same for all observations

- In many cases, the potential mean count varies systematically among subjects
  - Cars on a bridge: different times of day
  - Weeds in plots: different herbicide treatments, soil characteristics
  - Moles: race, age, sun exposure
- When there are explanatory variables that might relate to the changes in the mean count, we can use a POISSON REGRESSION MODEL
- We measure a response variable Y and explanatory variables $x_1, \ldots, x_p$
- The Poisson regression model treats $Y_i \sim Po(\mu_i)$, $i = 1, \ldots, n$, where

$$\log(\mu_i) = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip}$$

  - This is a form of GENERALIZED LINEAR MODEL with a Poisson random component and a log link
  - Using $\log(\mu)$ results in

$$\mu_i = \exp(\beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip})$$

    * This ensures that the mean count remains > 0 for any combination of explanatory variables but places no particular upper limit on the counts
    * Does create a curved relationship
    * Exponentially increasing or decreasing in x_j depending on the sign of β_j (see Figure 1)

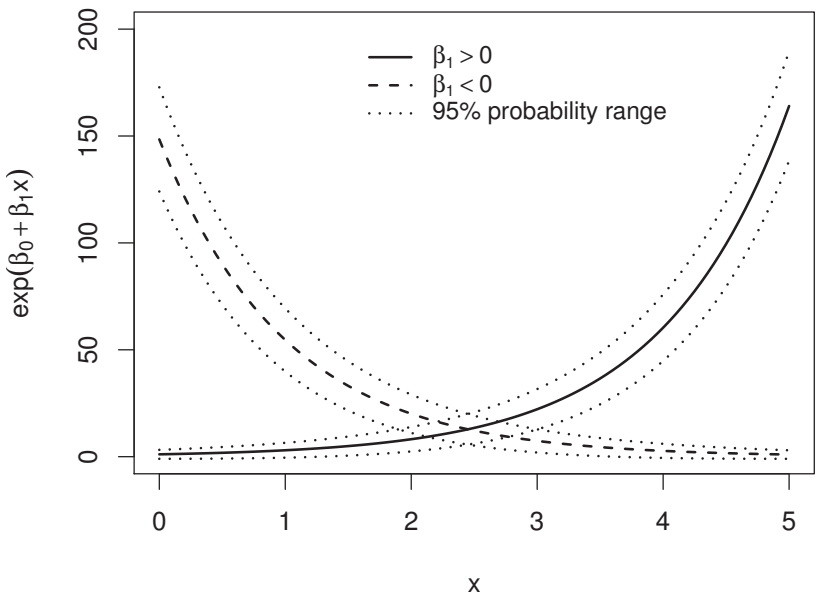
Interpretation of parameters in Poisson model

- $\log(\mu_i) = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip}$ or $\mu_i = \exp(\beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip})$
  - β_0 is the log mean of Y when all $x_j = 0$
  - β_j is the change in *log-mean* when x_j increases by 1 unit, holding other variables constant
    * To show this, we take differences of linear predictors again!
    * $\log(\mu_{x_1+1}) - \log(\mu_{x_1}) =$
      $[\beta_0 + \beta_1(x_1 + 1) + \beta_2 x_2 \ldots + \beta_p x_p] - [\beta_0 + \beta_1 x_1 + \beta_2 x_2 \ldots + \beta_p x_p] = \beta_1$
    * $\exp(\beta_j)$ *is the ratio of means* at $x_j + 1$ vs. x_j, holding other variables constant
- *We usually express changes in the Poisson mean as ratios or percentage changes rather than as differences*
  - Notice how smoothly the math works out for *logs of ratios*:

$$\log\left(\frac{\mu_1}{\mu_2}\right) = \log(\mu_1) - \log(\mu_2)$$

    which is just a difference of two linear predictors

Figure 1: Prototypes of Poisson Regression Curves (Model: $\log(\mu) = \beta_0 + \beta_1 x$)



- Also, in Poisson, larger mean implies larger variance as well since $Var(Y_i) = \mu_i$
  - A given size of difference has much larger variance if it was computed on larger means than on smaller means
  - For a given sample size, a certain difference between large means (say, 100 and 110) might be negligible (10% increase, $(110/100 - 1 = 0.1)$), while the same difference between small means (say, 11 and 1) might be hugely important (1000% increase $(11/1 - 1 = 10)$).
  - Equal differences in log-means (i.e., log-ratios) have a equal variance in Poisson model
    - Ratio estimate that equals $\log(110/100)$ has same variance as ratio estimate that equals $\log(1.1/1)$

- The *percentage* change in mean associated with a c-unit increase in x_j is $PC_j(c) = 100(\exp(c\beta_j) - 1)$
  - We often like to think in terms of percentage changes anyway

2.1 Parameter estimation and inference

Parameters are estimated by ML estimation, as usual.

- Observed counts $y_1, y_2, \ldots, y_n$ with corresponding explanatory variable measurements

- As always, we assume that the counts are independent of one another so that we can multiply PMFs to form a likelihood

- Log likelihood:

$$\begin{aligned}\log[L(\beta_0,\ldots,\beta_p|y_1,\ldots,y_n)] &= \sum_{i=1}^{n} \; [-\exp(\beta_0+\beta_1 x_{i1}+\ldots+\beta_p x_{ip}) \\ &\quad +y_i(\beta_0+\beta_1 x_{i1}+\ldots+\beta_p x_{ip}) \\ &\quad -\log(y_i!)]\end{aligned}$$

  - Generally no closed-form solution, so parameters are estimated using iterative numerical techniques
  - Resulting parameter estimates $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$
    * Corresponding standard errors $\sqrt{\widehat{Var}(\hat{\beta}_j)}$, $j = 0, 1, \ldots, p$ estimated in the "usual way" from the likelihood (See App. B.3.4)

- Estimated means (predicted or fitted values):

$$\hat{\mu} = \exp(\hat{\beta}_0 + \hat{\beta}_1 x_1 + \ldots + \hat{\beta}_p x_p)$$

- Estimated ratios of means
  - Estimated ratio of means at $x_j + c$ vs. x_j, holding other variables constant:

$$\exp(c\hat{\beta}_j)$$

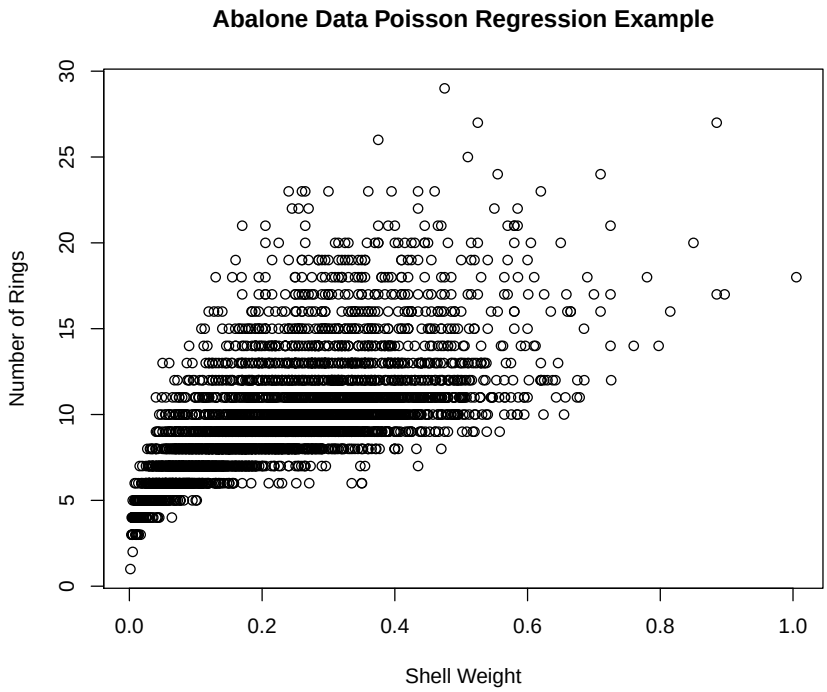
  - Expressed as a percentage change:

$$\widehat{PC}_j(c) = 100(\exp(c\hat{\beta}_j) - 1)$$

- Tests and confidence intervals *can* be done by Wald, although LR is preferred where available

- In R, Poisson regression model is fit in `glm(..., family=poisson(link="log"),...)`
  - `Anova()` from `car` does partial LR tests
    * `anova()` does sequential tests and model comparisons.
  - `confint()` does profile LR for individual parameters
  - LR inference on linear combinations of regression parameters is available using `mcprofile()`
    * Most things we want to estimate are linear combinations of regression parameters
    * So we have to do that thing with the coefficients again!
  - `emmeans()` can do Wald inference more automatically
    * Especially useful with categorical explanatories (later)

Figure 2: Abalone Regression data: `Rings` vs. `Shell`



<u>Example</u>: Age of Abalone (Lecture 19 scripts.R, abalone.csv)

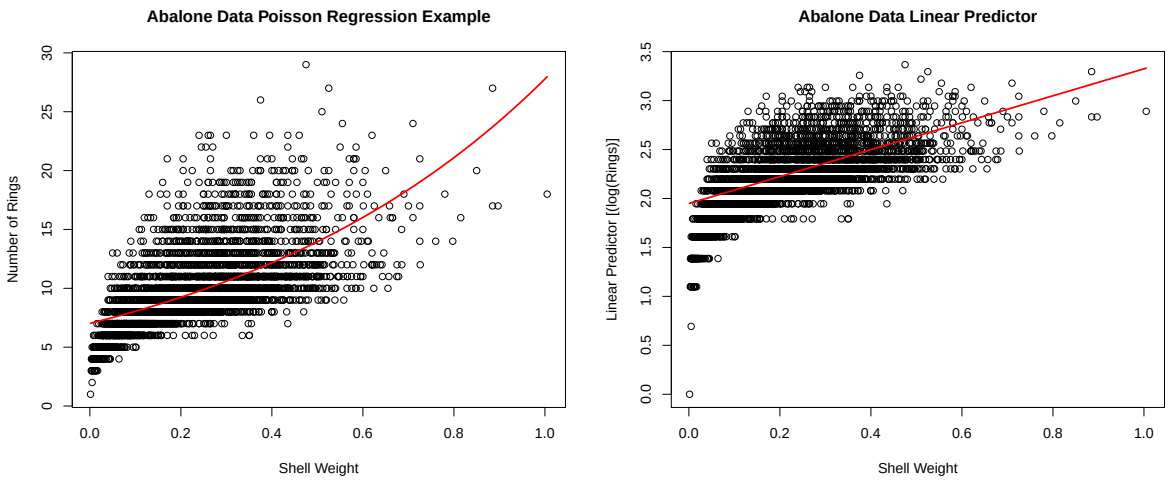
To keep things simple and to allow us to explore the data in a little more detail, we will model $Y =$<u>the number of rings</u> against $x =$<u>the shell weight</u>. We will explore the fit of the model and perform the usual tests and confidence intervals.

First, we should take a look at the data. Figure 2 shows a plot of `Rings` vs. `Shell`. We see a curved increasing relationship where the variance increases as the mean increases. If we fit `Rings`$\sim Po(\mu)$ with $\log(\mu) = \beta_0 + \beta_1$`Shell`, we would expect a positive "slope" parameter, β_1. However, given the shapes that $\exp(\beta_0 + \beta_1 x)$ can take on, we already see trouble. We therefore expect that will need to do something more than this simple model.

We will first fit that model and then show that the fit is poor:

```
> mod.fit <- glm(Rings~Shell, family=poisson(link="log"), data=aba)
> summary(mod.fit)

Call: glm(formula = Rings ~ Shell, family = poisson(link = "log"),
     data = aba)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
  -2.8566   -0.5270   -0.1866    0.3106    3.6570

Coefficients:
```

Figure 3: Abalone Poisson regression fit in the scale of the data (left) and in the scale of the linear predictor (right)




```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.94815    0.01013   192.3   <2e-16 ***
Shell        1.37619    0.03316    41.5   <2e-16 ***
---
Signif. codes:   0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

   Null deviance: 4139.3  on 4176  degrees of freedom
Residual deviance: 2487.4  on 4175  degrees of freedom
AIC: 19626

Number of Fisher Scoring iterations: 4
```

The estimated model is $\log(\mu) = 1.95 + 1.38\texttt{Shell}$, so we do indeed see a positive relationship between `Rings` and `Shell`. The Wald tests, which are probably just fine here because we have over 4000 observations, show that the parameter on `Shell` is highly significant, so there is clearly a relationship here. Of course, that doesn't mean that we have fit *the right* relationship. The model plot is shown in the left side of Figure 3. We see that the fit is not good. The curvature in the fitted mean is in completely the wrong direction. Since we have fit $\log(\mu) = \beta_0 + \beta_1 x$, we expect the log mean to be a straight line. If we plot the data in the $\log(\texttt{Rings})$ scale on the y-axis, the data should show *approximately* a straight line.[1] On the right side of Figure 3 is a plot in the log scale with the linear predictor overlaid. The points

[1]The mean of $\log(\texttt{Rings})$ is not exactly the same as the log of the mean of `Rings` because log is a nonlinear transformation, but the approximation is close enough for us to see major deviations. By the way, this works with logistic regression too. If one plots the data "in the logit scale"—meaning transform the observed proportions w_i/n_i using $\log[(w_i/n_i)/((n_i - w_i)/n_i)] = \log[(w_i/(n_i - w_i))]$, the log of the estimated odds—then the model fit from $\text{logit}(\pi) = \beta_0 + \beta_1 x$ is roughly a straight line. Note that this logit transformation won't

in the log scale show very sharp curvature, indicating a very poor fit. A better fit might be found if we use a transformation of x that has this shape.

---

3 Extending the model: Transformations and interactions

- Just as with linear and logistic regression, we may need to add transformations and/or interactions of explanatory variables to Poisson regression models.
  - We certainly need to do *something* with our current abalone model!
- **Transformations** operate the same way in Poisson regression as in other models.
  - Create a new variable of the form $z = g(x)$ for some function $g(\cdot)$
  - Logs, square roots, squares, and other functions are typical
    - Try to find function that has roughly the same shape as the data in the linear-predictor scale.
  - Interpreting effects of transformed variables faces same complication as in logistic regression
  - Parameter β_1 in $\log(\mu) = \beta_0 + \beta_1 z$ represents change in $\log(\mu)$ for a 1-unit change in z, not in x.
  - Change in $\log(\mu)$ for 1-unit change in x requires taking differences in linear predictor
    - $\log(\mu_{x+1}) - \log(\mu_x) = [\beta_0 + \beta_1 g(x+1)] - [\beta_0 + \beta_1 g(x)] = \beta_1[g(x+1) - g(x)]$
    - Different for each value of x
    - Could repeat this for a c-unit change
- **Polynomials** are a common tool for adding curvature into linear predictors
  - Simplest example is second-order (quadratic) model $\log(\mu) = \beta_0 + \beta_1 x + \beta_2 x^2$
  - We follow the same procedure to understand how a c-unit change affects the log-mean:

$$\begin{aligned}\log(\mu_{x+c}) - \log(\mu_x) &= [\beta_0 + \beta_1(x+c) + \beta_2(x+c)^2] - [\beta_0 + \beta_1(x) + \beta_2(x)^2] \\ &= \beta_1 c + \beta_2(2cx + c^2)\end{aligned}$$

  - Sort of a "blunt instrument" for modeling complicated shapes

---

work on binary data or on proportions that are 0 or 1, so it can sometimes be a little less informative to make this plot for logistic regression. It would work for only part of the placekick data, for example. See Section 5.2 in the book or later lectures for more details on assessing model fit.

    - Like any function, x^2 has a specific shape, which may not match the shape of the curvature in the model
    - In particular, quadratic models bend back in the opposite direction after a while, which may be unrealistic.
    - But is easy and adds some flexibility in shape compared to flat line.
        - "All models are wrong" but it might be better than a straight line

- **Interactions** allow the effects of one variable on the log-mean to change depending on the value of another variable.
  - Add cross-product term to model, e.g., $\log(\mu) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$
  - AGAIN, find total effect of each variable through differences in the linear predictors.
    - YOU do this. I'm tired.

<u>Example</u>: Age of Abalone (Lecture 19 scripts.R, abalone.csv)

We need to add curvature into the linear predictor. I show both the polynomial approach and some transformations.

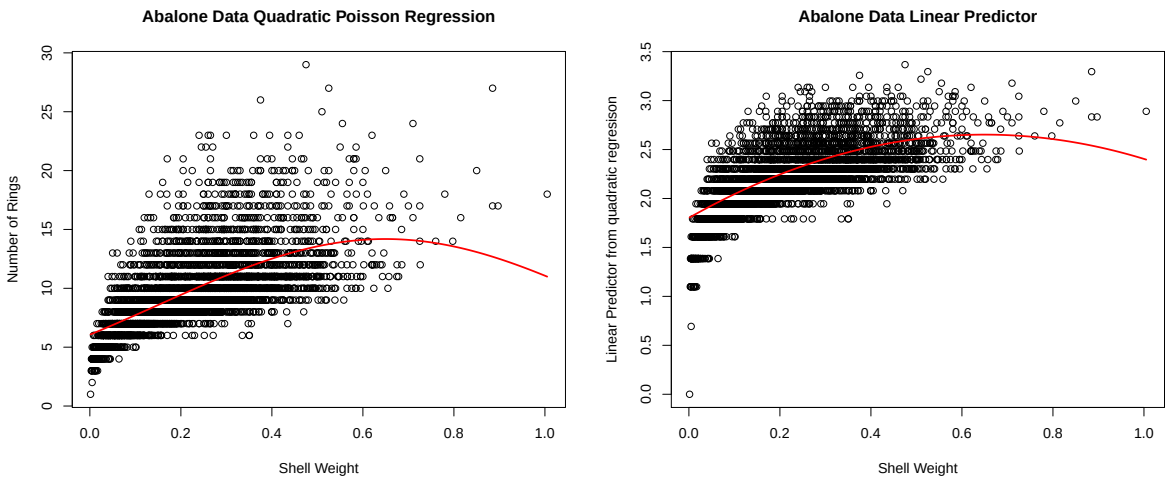
So first we try the quadratic model, $\log(\mu) = \beta_0 + \beta_1 \texttt{Shell} + \beta_2 \texttt{Shell}^2$:

```
> mod.fit2 <- glm(Rings~Shell + I(Shell^2),
                  family=poisson(link="log"), data=aba)
> summary(mod.fit2)$coefficients
             Estimate Std. Error   z value      Pr(>|z|)
(Intercept)  1.802195 0.01576838 114.29171  0.000000e+00
Shell        2.618719 0.10678935  24.52229 8.545969e-133
I(Shell^2)  -2.016550 0.16674115 -12.09389  1.137635e-33
```

The fitted model is $\log(\mu) = 1.80 + 2.62\texttt{Shell} - 2.02\texttt{Shell}^2$. The negative coefficient on the quadratic term means that the linear predictor is curving down, as we hoped it would, and it is highly significant, indicating that the model is a vast improvement over the linear-only version (the LR test in the program confirms this). See Figure 4 for plots of the fit. The fit is definitely better, but still not great around the ends. In particular, I don't like the fact that the fitted mean eventually starts to decrease with increasing shell weight. That seems unrealistic to me.

For the sake of demonstration, let's study this model further, even though we don't love it. We can estimate how much the mean number of rings changes as shell weight increases. I would choose, say, c =0.1 subjectively based on looking at the range of `Shell`, from 0 to 1. Notice that this depends on the weight, x, at which the comparison is made. The ratio of means at $x+0.1$ and x is $\exp[0.1*(2.62-2.02(2x+0.1))]$. In the program, plugging in some numbers as an example, we show how to compute this ratio and corresponding confidence intervals at initial shell weights of x =0.1, 0.2, and 0.3. The trick for LR inference, as usual, is to be able to properly identify the coefficients for the linear combinations of regression

Figure 4: Abalone Poisson **quadratic** regression fit in the scale of the data (left) and in the scale of the linear predictor (right)



parameters used to construct the estimates and intervals. You need to be able to write the quantity you are estimating in the form $a_0\beta_0 + a_1\beta_1 + \ldots + a_p\beta_p$, and then input the a_j coefficients into the `R` program.

Last, we fit the model using log(`Shell`) as the explanatory variable. The abridged output is below:

```
> mod.fitl <- glm(Rings~log(Shell), family=poisson(link="log"),
                                        data=aba)
> summary(mod.fitl)

Call:
glm(formula = Rings ~ log(Shell), family = poisson(link = "log"),
    data = aba)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.8269  -0.5138  -0.1995   0.2938   3.8928

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 2.767376   0.011766  235.21   <2e-16 ***
log(Shell)  0.297648   0.007105   41.89   <2e-16 ***
```

The fitted model is $\log(\mu) = 2.77 + 0.30\log(\texttt{Shell})$. Figure 5 shows that the fit appears to be *much* better now. It follows the sharper part of the concave bend very well, and it flattens rather than bending downward in large shells.

Another technique that is helpful is to plot the log-response against the transformed variable, which is approximately what our model expects to have a linear relationship. In

Figure 5: Abalone Poisson regression vs $\log(x)$ fit in the scale of the data (left) and in the scale of the linear predictor (right)

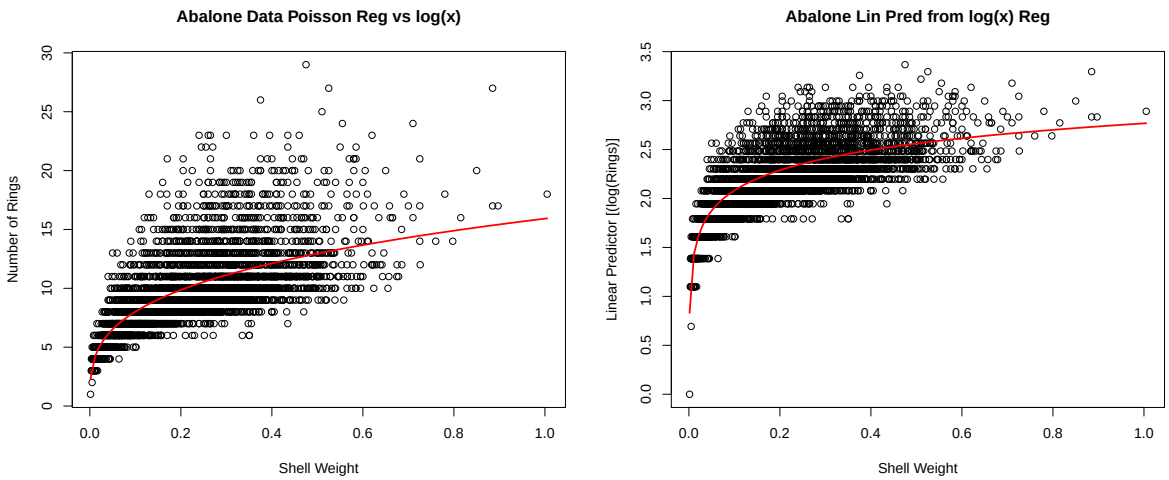


Figure 6, we plot log(`Rings`) and the linear predictor vs. log(`Shell`). If this transformation is correct, the plot should show a rather linear trend that the linear predictor cuts through. That's pretty much what we see—the linear predictor (red) cutting straight through a fairly linear-looking cloud of points. For reference, I've added a "lowess" smoother onto the plot (blue), which is just a kind of sliding average of the data. This gives us a graphical impression of the actual behaviour of the mean of log(`Rings`) across values of log(`Shell`). The curve barely deviates from the red straight line, suggesting that the fit of this model for the mean is pretty good, except possibly for the few very largest abalone.

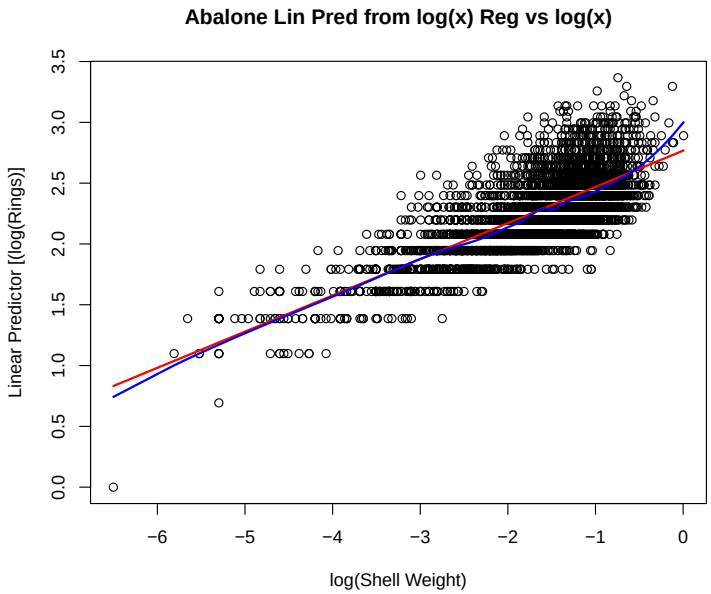
We could follow up with inferences on means and mean ratios as we did in the previous example. This is left as an exercise.

- The **Alcohol Consumption** example in the book fits a model with an interaction in it if you want to see an example of that process.
  - It's the same process we used for interactions in logistic regression, with interpretations being in terms of log-means rather than log-odds.

4 Conclusions: What to learn

1. The Poisson regression model is similar to the logistic regression model, and all of the things we have learned about logistic regression have parallels in Poisson regression.
   (a) Use what you know about logistic regression to make Poisson regression easier
   (b) Use what you learn about Poisson regression to reinforce or improve what you know about logistic regression

Figure 6: Replotting the linear predictor vs. log(x), since this should be approximately a straight line. Red = estimated model fit, Blue = LOWESS smoother.



2. In particular, be good at working with the linear predictors to figure out what you want to know

   (a) *Poisson just replaces the mean for the odds* in the systematic component of the GLM

      i. Odds ratio = exp(parameters) in logistic

      ii. Mean ratio = exp(parameters) in Poisson

3. Both are fit with `glm()`, so all of the estimation and inference tools from logistic regression apply here.

   (a) If you have to answer a question about a Poisson regression problem and you haven't been given code to do this, refer to your logistic regression code for ideas!

5 Exercises (due when announced)

1. In the Abalone example above, we fit $\log(\mu) = \beta_0 + \beta_1 \log(x)$. Another transformation with that same bent shape is $\sqrt{x}$. Fit a model with $\log(\mu) = \beta_0 + \beta_1\sqrt{x}$ to the data.

   (a) Plot the fit on top of the data using both the `Rings` scale and the log(`Rings`) scale on the Y axis, and both the `Shell` scale and the $\sqrt{\texttt{Shell}}$ scale on the X axis (in other words, 4 different plots).

      i. In which plot do we expect the linear predictor to be a straight line?

      ii. Comment on the fit of the model using all four plots: does the fit seem worse, similar, or better compared to the original x and the log?

   (b) Estimate the *mean number of rings* at shell weights of 0.2, 0.4, 0.6, and 0.8. Find the corresponding LR 95% confidence intervals. Check your results against the plot in Figure 5. Report the coefficient matrix along with the estimated means and confidence intervals.

   (c) Estimate the *percentage change in mean* for a 0.2-unit increase in shell weight based on starting at shell weights of 0.2, 0.4, and 0.6, and find the corresponding confidence intervals. Report the coefficient matrix along with the estimated percentage changes and confidence intervals. Do the ratios follow the trend you'd expect to see from the plots?

   (d) Repeat parts (b) and (c) with 95% Wald confidence intervals using `emmeans()` and just report the intervals.

2. Refer to the salamander data we used in Lecture 18. Use a Poisson regression to model the number of salamanders against the number of years since the last burn.

   (a) Plot the data twice, once in the original units and once using the log of the response. Comment on the results: does it look like a linear model in years will be a reasonably good fit?

   (b) Fit the Poisson regression model with years in linear form. Report the fitted model. Is the estimated mean trend upward or downward? Test whether this trend is significant and comment on the results.

   (c) Add a square term to the model. Report the fitted model. Test significance of the quadratic term. Does it suggest that there is important curvature in linear predictor versus age?

   (d) Use the *quadratic* model to estimate the percent change in mean number of salamanders with each 5-year increase in years since burn. Compute these changes starting from 0, 10, 20, and 30 years. Relate the pattern you to the sign of the estimated coefficient on the square term in the model.

   (e) Compute and report 95% LR confidence intervals for these percent changes.

STATISTICS 475: Applied Discrete Data Analysis

Lecture 20: Poisson Regression Models—Categorical Explanatory Variables

(B&L Section 4.2.3)

1 Problem to be solved

- We have seen how to write a Poisson regression model with numerical variables
- Now need to extend it to categorical variables and interactions
  - The methods are essentially exactly what you have done before for logistic regression (see Lecture 11)
  - We use the linear predictor to estimate log means and log mean ratios instead of log odds and log odds ratios

2 Categorical Variables

- In addition to a response Y, suppose X is categorical with q levels
  - We can create $q-1$ binary indicator (dummy) variables to represent the categories
    - R will use $x_2, \ldots, x_q$
    - Each $x_j,\ j = 2, \ldots, q$ is 1 if the observed value of x is the jth level, and 0 if not
    - "Indicates" the jth level
- Poisson regression model of Y against X:

$$Y \sim Po(\mu),\ \ \log(\mu) = \beta_0 + \beta_2 x_2 + \ldots + \beta_q x_q$$

  - Produced by `glm()` using formula `Y~X` when `X` is a factor-class variable

- As always, it helps to write out a table (here for $q = 4$ levels):

	Indicator Variable				
x-level	x_1	x_2	x_3	x_4	$\log(\mu_j) = \beta_0 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$
A	1	0	0	0	$\log(\mu_1) = \beta_0$
B	0	1	0	0	$\log(\mu_2) = \beta_0 + \beta_2$
C	0	0	1	0	$\log(\mu_3) = \beta_0 + \beta_3$
D	0	0	0	1	$\log(\mu_4) = \beta_0 + \beta_4$

  - From this, you can interpret the parameters:
    * β_0 is the log-mean of Y at the first level of x
      · This is the "baseline" level against which others are compared.
    * For $j = 2, \ldots, q$, $\beta_j = \log(\mu_j) - \log(\mu_1) = \log(\mu_j/\mu_1)$, the log-ratio of means between the jth group and the baseline group
    * For convenience of notation, we can sometimes pretend that there is a $\beta_1 x_1$ in the model, where $\beta_1 = 0$

- Any mean is estimated by exponentiating the right combination of parameters,

$$\hat{\mu}_j = \exp(\hat{\beta}_0 + \hat{\beta}_j)$$

2.1 Ratios of means

- Note: *everything* here is just the same as in logistic regression, where we compared probabilities through ratios of odds (ORs)

- *We compare means through their ratios, μ_a/μ_b for any two categories a and b*

- We already know that $\beta_j = \log(\mu_j/\mu_1)$, so

$$\frac{\hat{\mu}_j}{\hat{\mu}_1} = \exp(\hat{\beta}_j)$$

- For other pairs, work through the logs:

$$\log(\mu_a/\mu_b) = \log(\mu_a) - \log(\mu_b) = [\beta_0 + \beta_a] - [\beta_0 + \beta_b] = \beta_a - \beta_b$$

  - So

$$\frac{\hat{\mu}_a}{\hat{\mu}_b} = \exp(\hat{\beta}_a - \hat{\beta}_b)$$

  - This can be treated as the general formula for any two means, where we understand that $\beta_1 = 0$
  - *So ratios of means are just exponentiated differences between the numerator- and denominator-level linear predictors*

- We already know how to do Wald and LR inferences on differences of regression parameters that are estimated by MLEs
  - Estimate $\log(\mu_a/\mu_b) = \beta_a - \beta_b$with $\hat{\beta}_a - \hat{\beta}_b$, the MLE of the parameter difference
  - Approximately normally distributed in large samples
  - Test of equal means uses Wald or LR test, with H_0 : ratio = 1, or equivalently $H_0 : \beta_a - \beta_b = 0$
  - Get $\sqrt{\widehat{Var}(\hat{\beta}_a - \hat{\beta}_b)}$ mathematically (using the formula for variances of linear combinations) and computationally for Wald
    * `emmeans()` does this automatically with `specs=~X` when `X` is a factor
  - LR CI through `mcprofile` uses a coefficient matrix where each row contains 1 in the ath position, -1 in the bth position, and a 0 everywhere else
    * If either a or b is 1, then need no coefficient for that parameter
  - Exponentiate endpoints of the resulting confidence intervals
- Very often test for a "variable X effect": do *any* of the levels of X have different means?
  - $H_0 : \mu_1 = \mu_2 = \ldots = \mu_q$
  - $H_0 : \beta_2 = \beta_3 = \ldots = \beta_q = 0$ versus $H_a :$ Not all $\beta_j = 0$
  - LR model comparison test
    * How many df?

<u>Example</u>: Age of Abalone (Lecture 20 scripts.R, abalone.csv)

There is one categorical variable in the abalone data: `Sex` with **three levels**! It is currently stored as a numerical variable—0=infant, 1=male, 2=female—so the first thing we will need to do is convert it to a factor, say `Sexf`. We will start by fitting a simple model with just `Sexf` as the only explanatory variable. Later we will add numeric variables and interactions.

When the factor variable `Sexf` is created, the levels are stored in the original numeric order. (We could rearrange the order of the levels using the if we wanted to.) Thus, when we start with a model that has only `Sexf` in the model, we will need to create indicators for the last two levels of `Sexf`, say x_M and x_F. Then the model will be $\log(\mu) = \beta_0 + \beta_M^S x_M + \beta_F^S x_F$ where the ^{S} is just to name the factor that the parameter belongs to (`Sexf`) and the $_M$ and $_F$ are the levels. You should get good at filling out the table for indicators, parameters, and means:

	Indicator Variable		
Sex	x_M	x_F	$\log(\mu_j) = \beta_0 + \beta_M^S x_M + \beta_F^S x_F$
I			$\log(\mu_1) =$
M			$\log(\mu_2) =$
F			$\log(\mu_3) =$

The model fit is below:

```
> aba$Sexf <- factor(x=aba$Sex, labels=c("I", "M", "F"))
> head(aba[,c("Sex","Sexf","Shell","Rings")])
  Sex Sexf Shell Rings
1   1    M 0.150    15
2   1    M 0.070     7
3   2    F 0.210     9
4   1    M 0.155    10
5   0    I 0.055     7
6   0    I 0.120     8
>
> # First model: Rings vs. Sex.
> # Fit model, estimate means, explore comparisons among sexes
> mod.fit1 <- glm(Rings~Sexf, family=poisson(link="log"), data=aba)
> summary(mod.fit1)

Call: glm(formula = Rings ~ Sexf, family = poisson(link = "log"),
    data = aba)
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-3.1064  -0.6604  -0.2181   0.3862   4.4504

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 2.065655   0.009718  212.56   <2e-16 ***
SexfM       0.305103   0.012473   24.46   <2e-16 ***
SexfF       0.343927   0.012774   26.92   <2e-16 ***
--- Signif. codes:   0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 4139.3  on 4176  degrees of freedom
Residual deviance: 3261.8  on 4174  degrees of freedom
AIC: 20402

Number of Fisher Scoring iterations: 4
>
> library(car)
> Anova(mod.fit1)
Analysis of Deviance Table (Type II tests)

Response: Rings
     LR Chisq Df Pr(>Chisq)
Sexf   877.51  2  < 2.2e-16 ***
```

The fitted model gives $\log(\mu) = 2.07 + 0.31x_M + 0.34x_F$. The LR test shows that the sex

factor is highly significant, meaning that not all sexes have the same mean number of rings.

We next estimate each sex mean with a confidence interval. Details for computing LR intervals are in the program, but it is trivially easy to get Wald intervals using `emmeans()`:

```
> emm1 = emmeans(mod.fit1, specs=~Sexf)
> confint(emm1, type="response")
 Sexf  rate     SE  df asymp.LCL asymp.UCL
 I     7.89 0.0767 Inf      7.74      8.04
 M    10.71 0.0837 Inf     10.54     10.87
 F    11.13 0.0923 Inf     10.95     11.31

Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

The mean number of rings for infants is $e^{\hat{\beta}_0}$ =7.89 with a 95% LR confidence interval from 7.71 to 8.07. This is considerably below the 95% confidence intervals for the mean number of rings for males (10.51–10.91) and for females (10.91–11.35), so clearly the infants have a smaller mean number of rings than the other abalone. This is to be expected, although it is a bit of a surprise that they are not much smaller. I suspect that that "I" might really stand tor "Immature" or even "Indeterminate", meaning that the sex could not be determined, either because the animal was too young or the coloring was just not obviously either green or beige.

Next, we compare pairs of means by making ratios: $\mu_M/\mu_I, \mu_F/\mu_I$, and μ_M/μ_F. The hardest part of this is figuring out which argument values to use for the `method` argument `contrast(...,method=,...)`. You can try to remember that the pairwise differences in the linear predictors (logged ratios of means) are all computed in the same order as the factor levels are stored. Using `pairwise` uses earlier levels in numerators, and `revpairwise` uses earlier levels in denominators. Or you can do what I did and try both to see what gives me what I want... :) The results are below.

```
> confint(contrast(emm1, method="pairwise"), type = "response")
 contrast ratio      SE  df asymp.LCL asymp.UCL
 I / M    0.737 0.00919 Inf     0.716     0.759
 I / F    0.709 0.00906 Inf     0.688     0.731
 M / F    0.962 0.01096 Inf     0.937     0.988

Confidence level used: 0.95
Conf-level adjustment: tukey method for comparing a family of 3 estima
Intervals are back-transformed from the log scale
> confint(contrast(emm1, method="revpairwise"), type = "response")
 contrast ratio     SE  df asymp.LCL asymp.UCL
 M / I     1.36 0.0169 Inf      1.32      1.40
 F / I     1.41 0.0180 Inf      1.37      1.45
 F / M     1.04 0.0118 Inf      1.01      1.07
```

```
Confidence level used: 0.95
Conf-level adjustment: tukey method for comparing a family of 3 estima
Intervals are back-transformed from the log scale
```

Clearly, the ratios of means for M:I and F:I are larger than 1. Males have about 36% more rings on average than infants, and females have 41% more. Interestingly, the comparison of males to females shows that males have fewer rings on average than females (mean ratio is 0.962, with 95% Wald CI 0.937–0.988)! I wonder why that would be...maybe females tend to live longer, and so more of them in the sample have larger numbers of rings??? Or perhaps just luck of the sample, since the size of the effect is pretty small—a 4% difference—and the upper boundary is not far from 1 relative to the width of the interval.

The R script also shows how to get the LR intervals. They are very similar to these when comparable error-rate adjustments are used..

---

3 Interactions between categorical and numerical variables

- Interactions between categorical and numerical variables are straightforward to handle
- To start, suppose we have a numeric variable Z to accompany our categorical explanatory X.
  - A model can include Z, indicators for X, and optionally the interactions between X and Z.
    - Interactions are represented by cross-products of Z with the X indicators.
    - $Y \sim Po(\mu)$, where $\log(\mu) = \beta_0 + \beta_2^X x_2 + \ldots + \beta_q^X x_q + \beta^Z z + \beta_2^{XZ} x_2 z + \ldots + \beta_q^{XZ} x_q z$
  - As always, it helps to write out a table (here depicting $q = 4$ levels for X):

	Indicators				$\log(\mu) = \beta_0 + \beta_2^X x_2 + \beta_3^X x_3 + \beta_4^X x_4$
x-level	x_2	x_3	x_4	Z	$+\beta^Z z + \beta_2^{XZ} x_2 z + \beta_3^{XZ} x_3 z + \beta_4^{XZ} x_4 z$
A	0	0	0	z	$\log(\mu_{1,z}) = \beta_0 + \beta^Z z$
B	1	0	0	z	$\log(\mu_{2,z}) = \beta_0 + \beta_2^X + \beta^Z z + \beta_2^{XZ} z = (\beta_0 + \beta_2^X) + (\beta^Z + \beta_2^{XZ})z$
C	0	1	0	z	$\log(\mu_{3,z}) = \beta_0 + \beta_3^X + \beta^Z z + \beta_3^{XZ} z = (\beta_0 + \beta_3^X) + (\beta^Z + \beta_3^{XZ})z$
D	0	0	1	z	$\log(\mu_{4,z}) = \beta_0 + \beta_4^X + \beta^Z z + \beta_4^{XZ} z = (\beta_0 + \beta_4^X) + (\beta^Z + \beta_4^{XZ})z$

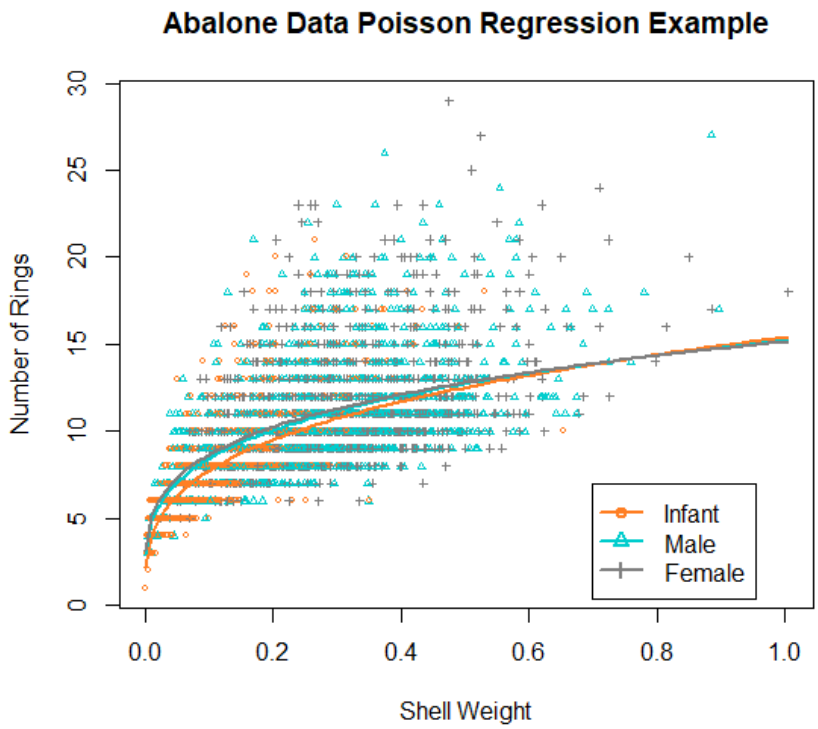
  - Notice that this simply creates a separate straight line—(intercept) + (slope)*z—for each level of X
  - Interpret each parameter: what does it measure?

Example: Age of Abalone (Lecture 20 scripts.R, abalone.csv)

The next model fit involves sex, shell weight, and the interaction. Because we found previously that the log of the shell weight fits the data fairly well, we use the model $\log(\mu) = \beta_0 + \beta_M^S x_M + \beta_F^S x_F + \beta^W \log(w) + \beta_M^{SW} x_M \log(w) + \beta_F^{SW} x_F \log(w)$, where w represents the shell **w**eight ($\log(w)$ is serving the role of z in the table above). I leave it to you to create the table of log means and parameters if you wish. The data are plotted in Figure 1, broken out by sex. We fit the model with interactions, print the summary, and perform LR tests for each effect.

```
> mod.fit2 <- glm(Rings~Sexf*log(Shell), family=poisson(link="log"),
                  data=aba)
> summary(mod.fit2)

Call: glm(formula = Rings ~ Sexf * log(Shell),
          family = poisson(link = "log"), data = aba)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.8171  -0.5036  -0.1848   0.2909   3.9309

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)       2.73370    0.02952  92.599   <2e-16 ***
SexfM            -0.01210    0.03586  -0.337   0.7358
SexfF            -0.01661    0.03838  -0.433   0.6653
log(Shell)        0.29908    0.01301  22.987   <2e-16 ***
SexfM:log(Shell) -0.04241    0.01925  -2.203   0.0276 *
SexfF:log(Shell) -0.05670    0.02266  -2.502   0.0124 *
--- Signif. codes:   0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

   Null deviance: 4139.3  on 4176  degrees of freedom
Residual deviance: 2151.4  on 4171  degrees of freedom
AIC: 19298

Number of Fisher Scoring iterations: 4
>
> Anova(mod.fit2)
Analysis of Deviance Table (Type II tests)
Response: Rings
                LR Chisq Df Pr(>Chisq)
Sexf               29.34  2  4.261e-07 ***
log(Shell)       1102.36  1  < 2.2e-16 ***
```

Figure 1: Abalone Poisson regression: Rings vs. Shell Weight, Sex, and the interaction




```
Sexf:log(Shell)      8.06   2     0.01774 *
```

The estimated model is $\log(\mu) = 2.73 - 0.012x_M - 0.017x_F + 0.30\log(w) - 0.042x_M\log(w) - 0.057x_F\log(w)$, and the fit is represented in the colour-coded plot in Figure 1. The LR test for the interaction effect tests $H_0 : \beta_M^{SW} = \beta_F^{SW} = 0$. The test statistic is 8.06 and the p-value is 0.018< 0.05, which suggests that the interaction is significant. The "slopes" in the linear predictor (the relationship between shell weight and number of rings) are not all the same for all three sexes.

The plot helps to interpret the parameters. For example, $\hat{\beta}_0 = 2.73$ is the mean log(`Rings`) when both `Sexf` variables are 0 (i.e., sex is Infant), and the log(`Shell`)=0 (i.e., `Shell`=1!). Also, when log(`Shell`)=0, males have a 0.012 *smaller* mean log(`Rings`) than infants—we do see that the orange line does cross above the others off the right end of the plot, but this is a considerable extrapolation beyond the last observed infant shell weight. For Infants, as log(`Shell`) increases by 1 unit, the log mean number of rings increases by 0.30. This is not the same as the change for a 1-unit increase in shell weight! (That is left as an exercise.) For males, the change in log(`Rings`) for a 1-unit change in log(`Shell`) is 0.042 *less* than it is for infants. This is seen in the plot from the orange line's slightly more rapid increase than the blue line over the course of the plot.

Estimating and comparing means is possible using either Wald (`emmeans`) or LR (`mcprofile`) methods. In particular, `emmeans()` will produce log-means for each level of any factor variables listed in `specs=` and at the mean values of numeric variables or, alternatively, levels

listed in `at=`. Below, we show the means at each level of `Sexf` at shell weights of 0.1, 0.2, and 0.3.[1] These means could be compared using some method within `contrast()`.

```
> emm2 = emmeans(mod.fit2, specs=~Sexf+ Shell,
              at=list(Shell=c(0.1, 0.2, 0.3)))
> confint(emm2, type="response")
 Sexf Shell  rate     SE  df asymp.LCL asymp.UCL
 I      0.1  7.73 0.0768 Inf      7.58      7.88
 M      0.1  8.42 0.1341 Inf      8.16      8.69
 F      0.1  8.66 0.1847 Inf      8.31      9.03
 I      0.2  9.51 0.1135 Inf      9.29      9.74
 M      0.2 10.06 0.0886 Inf      9.89     10.23
 F      0.2 10.25 0.1098 Inf     10.03     10.46
 I      0.3 10.74 0.1676 Inf     10.41     11.07
 M      0.3 11.16 0.0893 Inf     10.99     11.34
 F      0.3 11.31 0.0941 Inf     11.12     11.49

Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

Another interesting thing to do is compare *slopes* of the regression models across the levels of the factor. There is another function in this package, `emtrends()`, that can produce estimate of slopes and compare them. See the program for an example.

4 Conclusions: What to learn

1. Categorical explanatories are handled in exactly the same way in both Poisson and logistic regression.
   (a) The only change is that we are modeling a log-mean count response instead of a log-odds of success
   (b) Mean ratios instead of odds ratios, but processes are basically identical
2. New distribution doesn't change the theory—it's all from GLMs
3. Application is the same, too
   (a) `emmeans` and `mcprofile` work the same way with the same pros and cons

[1] In `glm()` our formula specified `log(Shell)`. So `glm()` looks for a variable called `Shell` and takes the log internally. Thus, in functions like `predict()` and `emmeans()` we need to provide values of `Shell`, not log(`Shell`). If we had instead created a new variable, for example using `aba$logshell <- log(aba$Shell)` we could have included `logshell` as a variable in `glm()` and fit the exact same model. Everything would be the same, except that now we would need to provide values of `logshell` to any functions that use the model to produce estimated linear predictors.

5 Exercises (due when announced)

Refer to the low birth weight data, where we used a Poisson regression to model the number of doctor visits (`ftv`) against the age of the woman. In Lecture 11, we created factors out of the variables `smoke`, `race`, `ht`, and `ui`. We will do the same thing here to see which variables might be related to how often a pregnant woman needed to visit a doctor.

1. In the first Abalone example above where we modeled `Rings` against `Sex` for the Abalone data, we compared means for all three `Sex` levels. Work out the math to derive these coefficients for each comparison.

   (a) μ_M/μ_I

   (b) μ_F/μ_I

   (c) μ_M/μ_F

2. In the second Abalone example above, we fit $\log(\mu) = \beta_0 + \beta_M^S x_M + \beta_F^S x_F + \beta^W \log(w) + \beta_M^{SW} x_M \log(w) + \beta_F^{SW} x_F \log(w)$.

   (a) Notice that, according to the x-axis of Figure 1, an increase of, say, 0.1 units of shell weight is a reasonable amount to Estimate the mean number of rings for each sex at shell weights of 0.1, 0.2, 0.3, and 0.4, and find the corresponding confidence intervals. Notice that these estimates should match visually with the lines in Figure 1. Report the coefficient matrix for `mcprofile` along with the estimated means and LR confidence intervals.

   (b) Estimate the percentage change in mean for a 0.1-unit increase in shell weight based on starting at shell weights of 0.1, 0.2, and 0.3 for each sex, and find the corresponding confidence intervals. Check your results against the answers you got in part (a)—manually do the divisions of the means from that part to confirm the percentage changes in this part. Report the coefficient matrix along with the estimated percentage changes and confidence intervals.

   (c) Estimate ratios of means between M:I, F:I, and M:F at shell weights of 0.2 and 0.3, and express these as percentage changes. Also find confidence intervals for these percentages. Check your results against the answers you got in part (a)—manually do the divisions of the means from that part to confirm the percentage changes in this part. Report the coefficient matrix along with the estimated percentage changes and confidence intervals.

3. Repeat Exercise 2 using Wald methods.

Extra exercise for practice

1. Suppose we have a problem with two categorical variables, A with three levels and B with two levels, and two numerical variables, $X$and Z.

   (a) Create the table of indicators using a's for the A indicators and b's for the B indicators. Include the numerical variables in the table.

(b) Consider a model where we expect X to interact with A and Z to interact with B. Add the logits to the table for each combination of A and B. Interpret the parameters.

(c) We want to compute separate mean ratios and LR confidence intervals corresponding to a 1 unit increase in X and for a 1-unit increase in Z. Figure out the coefficient matrix for `mcprofile`.

(d) Also determine the coefficient matrix for comparing the mean ratios for the three levels of A, 1:2, 1:3, and 2:3, holding X and Y fixed at specific values, x and y, respectively.

(e) Figure out the code sequence for repeating these analyses using Wald intervals instead

STATISTICS 475: Applied Discrete Data Analysis

Lecture 21: Poisson Loglinear Models

(B&L Section 4.2.4)

1 Problem to be solved

- We have seen how to write a Poisson regression model with numerical variables or one categorical variable
- Now need to extend it to two or more categorical variables
  - This allows us to model contingency tables in a different way that allows much greater flexibility than just doing a Pearson test and the meager follow-ups.
  - We will be able to handle more than two categorical variables
  - We will have a built-in systematic way of doing follow-up investigations on tables
- **This is probably the most common use for Poisson regression models.**

2 Poisson models for contingency tables: Loglinear models

- Now consider *two* categorical variables, X with I levels and Z with J levels.
  - This is just the same contingency table situation we had in Lecture 15
  - Changing notation slightly to mesh with notation for Poisson regression
- We assume that for each combination of X and Z (a "cell" in the contingency table), there is a random variable Y_{ij} $i = 1, \ldots, I$; $j = 1, \ldots, J$ (formerly N_{ij} in contingency tables).
- We further assume that each count is observed independently of each other
  - i.e., treating each (i, j) combination as having a separate count RV that is not influenced by the units that make up the count for any other Y

- Not to be confused with "independence" of variables X and Z as a null hypothesis for a test—we will get to that soon.
- Just means that adding more to one (i, j) count doesn't have to take away from other cells' potential counts
  - Separately accumulating counts in each cell; not conditioning on observing some fixed total as multinomial assumes

- Finally, we assume that each cell count $Y_{ij} \sim Po(\mu_{ij})$
  - Previously assumed multinomial counts
    - Assumed that total number of trials, n, or the row marginal counts were known in advance
    - Not always realistic that counts come from trials or that totals are fixed
    - In many studies the total sample size is not fixed in advance, but is random resulting from the total of the cell counts
      - Just spend a certain amount of *time* recording information, and observe whatever counts you get.
  - Now assuming totals or marginal counts are not necessarily known in advance.

- At each combination of X and Z observe a count, y_{ij}, $i = 1, \ldots, I$; $j = 1, \ldots, J$.
  - Represent cross-tabulation of counts in a CONTINGENCY TABLE
  - Have seen contingency tables before, but now use notation from a Poisson model, y_{ij} instead of n_{ij}

		Z				
		1	2	$\cdots$	J	Total
X	1	y_{11}	y_{12}	$\cdots$	y_{1J}	y_{1+}
	2	y_{21}	y_{22}	$\cdots$	y_{2J}	y_{2+}
	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	
	I	y_{I1}	y_{I2}	$\cdots$	y_{IJ}	y_{I+}
	Total	y_{+1}	y_{+2}	$\cdots$	y_{+J}	$y_{++} = n$

- Typical interests in the analysis of a contingency table:
  - Are X and Z are "independent"?
    - Independence means that the conditional distribution of X is the same for each level of Z and vice versa.
    - Expressed now through expected counts within each row rather than probabilities
      - In row i, column j we have μ_{ij} from each Poisson RV
    - Independence means that these means follow the same *proportional* pattern in every row

    * For example consider these μ_{ij}'s:

μ_{ij}		Z			
		1	2	3	Total
X	1	10	40	20	70
	2	7	28	14	49
	Total	17	68	34	119

    * Notice that each row follows the same 1:4:2 pattern of counts, as do the column marginal counts
    * Similarly columns are all in a 10:7 ratio, as is the margin
  - Sometimes equally important: describe the nature of any association

- Approaches to analysis
  - We have seen the "usual" Pearson test of independence, which is also the score test from a multinomial model
  - We mentioned the likelihood ratio test but did not focus on it
  - Now we will model the table using a Poisson *loglinear model*
    * A LOGLINEAR MODEL is just a Poisson regression model for 2 or more categorical explanatory variables
    * We will focus on modeling, so LR tests are natural here.

2.1 Poisson (loglinear) model for two categorical explanatory variables

- Start by assuming that the count for each combination of X and Z has a Poisson distribution with its own mean,

$$Y_{ij} \sim Po(\mu_{ij}),\ i = 1, \ldots, I;\ j = 1, \ldots, J.$$

- We want to model the means as functions of the explanatories, as we have been doing.
- Usual approach to categorical explanatories
  - Indicator variables $x_1, x_2, \ldots, , x_I$ for X and $z_1, z_2, \ldots, z_J$ for Z
  - R will drop first of each set
- Use together in a Poisson regression model with

$$\log(\mu) = \beta_0 + \beta_2^X x_2 + \beta_3^X x_3 + \ldots + \beta_I^X x_I + \beta_2^Z z_2 + \beta_3^Z z_3 + \ldots + \beta_J^Z z_J$$

- Simplest example is a 2×2 contingency table ($I = J = 2$)

- Need indicators for level 2 of X and level 2 of Z[1]

X	Z	x_2	z_2	$\log(\mu_{ij}) = \beta_0 + \beta_2^X x_2 + \beta_2^Z z_2$
1	1	0	0	$\log(\mu_{11}) = \beta_0$
1	2	0	1	$\log(\mu_{12}) = \beta_0 + \beta_2^Z$
2	1	1	0	$\log(\mu_{21}) = \beta_0 + \beta_2^X$
2	2	1	1	$\log(\mu_{22}) = \beta_0 + \beta_2^X + \beta_2^Z$

  * Abbreviate this model for convenience:

$$\log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z,\ i = 1, 2,\ j = 1, 2,$$

    where is it implicit that $\beta_1^X = \beta_1^Z = 0$

- β_0 is the log-mean count when all variables are set to their first levels
  * i.e., at combination (1,1)
- β_2^X measures the difference in log-means (i.e., the log of the ratio of means) between counts in rows 2 and 1 in the same column.
  * $\beta_2^X = \log(\mu_{21}) - \log(\mu_{11}) = \log(\mu_{22}) - \log(\mu_{12})$
    · Remember that $\log(a) - \log(b) = \log(a/b)$
  * Notice that this same difference or ratio holds in every column (for every level j of Y)
    · The effect on the log-means from changing X from level 2 to level 1 is the same regardless of which level of Z we work from
  * Hence, this model imposes independence between X and Z
- The same is would be true in a larger table: $\beta_i^X = \log(\mu_{i1}/\mu_{11}) = \log(\mu_{i2}/\mu_{12}) = \ldots = \log(\mu_{iJ}/\mu_{1J})$ for each i and for all levels j of Z
  * It can therefore be shown that these parameters also measure differences in log-means, or log-ratios of means, in the *margins* of the table: $\beta_i^X = \log(\mu_{i+}/\mu_{1+})$ for each i
- An analogous interpretation holds for each β_j^Z

Odds Ratios

- As suggested before, we can use odds ratios between any two rows and columns to describe the association within a contingency table
  - By analogy to binomial where $\mu = n\pi$, we can write ORs in terms of mean counts rather than probabilities

[1] In this table, as in all other cases with categorical explanatories, feel free to insert β_1^X and β_1^Z where it "looks like" they belong. It may help you to appreciate the simple structure of the log means better. You just have to remember to remove these parameters when you convert calculations on the linear predictors into estimates and confidence intervals in R.

- Think of a 2×2 table:

$$OR_{12,12} = \frac{\pi_{11}/(1-\pi_{11})}{\pi_{21}/(1-\pi_{21})} = \frac{\pi_{11}/\pi_{12}}{\pi_{21}/\pi_{22}} = \frac{n\pi_{11}/n\pi_{12}}{n\pi_{21}/n\pi_{22}} = \frac{\mu_{11}/\mu_{12}}{\mu_{21}/\mu_{22}}$$

  * Explains the entire association in the table
    - Numerator is ratio of means from column 1 and column 2 in the *first* row
    - Denominator is ratio of means from column 1 and column 2 in the *second* row
    - OR produces a comparison of these mean ratios
    - Can equivalently write it as ratio of mean ratio in column 1 to mean ratio in column 2

$$OR_{12,12} = \frac{\mu_{11}/\mu_{21}}{\mu_{12}/\mu_{22}}$$

- Either way be rearranged as "X-product" in the table

$$OR_{12,12} = \frac{\mu_{11}\mu_{22}}{\mu_{21}\mu_{12}}$$

- In a larger table, association may be more complex, involving numerous rows and columns
  - Hard to measure complicated association in one number
  - Instead, can analyze subsets of the table: pick pairs of rows and columns and look at association in pieces
  - For any two rows, i_1, i_2 and any two columns j_1, j_2 can compute the "local" odds ratio

$$OR_{i_1 i_2, j_1 j_2} = \frac{\mu_{i_1 j_1}\mu_{i_2 j_2}}{\mu_{i_1 j_2}\mu_{i_2 j_1}}$$

- Notice that $\log(OR_{i_1 i_2 \backslash j_1 j_2}) = \log(\mu_{i_1 j_1}) + \log(\mu_{i_2 j_2}) - \log(\mu_{i_1 j_2}) - \log(\mu_{i_2 j_1})$
  - So it is very convenient to write log-ORs in terms of model parameters by manipulating linear predictors
    * Same ideas as in logistic regression!
  - In the 2×2 table example, our model gives us

$$\begin{aligned} \log(OR_{12\backslash 12}) &= \log(\mu_{11}) + \log(\mu_{22}) - \log(\mu_{12}) - \log(\mu_{21}) \\ &= [\beta_0] + [\beta_0 + \beta_2^X + \beta_2^Z] - [\beta_0 + \beta_2^Z] - [\beta_0 + \beta_2^X] \\ &= 0! \end{aligned}$$

    * Thus, $OR = e^0 = 1$, meaning that there is no association between X and Z
      - i.e., X and Z are independent

- What happened?

- Nothing in this model allows the ratios of means between two rows to change across columns or vice versa.
- The model represents the means assuming independence between X and Z
  - Count the parameters in the model: $1 + (I-1) + (J-1)$, leaving $IJ - [1 + (I-1) + (J-1)] = (I-1)(J-1)$ residual degrees of freedom
  - Exactly the same as in a test for independence!
    - (This is no coincidence)
- Need to augment the model to allow these ratios to change

2.2 Loglinear model with interaction

- Adding cross-product interaction terms $x_i z_j$ as variables in the model allows the row mean-ratios differ among columns and vice versa
  - There are $(I-1)(J-1)$ such parameters
- Augmented model is

$$\log(\mu) = \beta_0 + \beta_2^X x_2 + \beta_3^X x_3 + \ldots + \beta_I^X x_I + \beta_2^Z z_2 + \beta_3^Z z_3 + \ldots + \beta_J^Z z_J + \beta_{22}^{XZ} x_2 z_2 + \ldots + \beta_{IJ}^{XZ} x_I z_J$$

  - This is a SATURATED MODEL: there are as many non-zero model parameters (IJ) as there are counts to be modeled
    - $1 + (I-1) + (J-1) + (I-1)(J-1) = IJ$
  - The estimated mean counts from this model equal the observed counts exactly
  - Residual deviance and residual DF are both 0
- Because of the way indicators work, pulling off one parameter from each variable, a convenient way to write the loglinear model is

$$\log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_{ij}^{XZ};\ i = 1, \ldots, I;\ j = 1, \ldots, J$$

  - Any parameter with index $i = 1$ or $j = 1$ is zero by definition.
- 2×2 example:

X	Z	x_2	z_2	Log-Mean
1	1	0	0	$\log(\mu_{11}) = \beta_0$
1	2	0	1	$\log(\mu_{12}) = \beta_0 \quad + \beta_2^Z$
2	1	1	0	$\log(\mu_{21}) = \beta_0 + \beta_2^X$
2	2	1	1	$\log(\mu_{22}) = \beta_0 + \beta_2^X + \beta_2^Z + \beta_{22}^{XZ}$

- Parameter interpretation:

- β_0 is the log-mean count when all variables are set to their first levels
  - i.e., at combination (1,1)
- β_2^X measures the difference in log-means (i.e., the log of the ratio of means) between counts in rows 2 and 1 *in column 1 only.*
  - $\beta_2^X = \log(\mu_{21}) - \log(\mu_{11}) \neq \log(\mu_{22}) - \log(\mu_{12}) = \beta_2^X + \beta_{22}^{XZ}$
- An analogous interpretation holds for β_j^Z: $\beta_j^Z = \log(\mu_{+j}/\mu_{+1})$
- From this we can see that

$$\beta_{22}^{XZ} = [\log(\mu_{22}) - \log(\mu_{12})] - [\log(\mu_{21}) - \log(\mu_{11})] = \log\left(\frac{\mu_{22}/\mu_{12}}{\mu_{21}/\mu_{11}}\right) = \log(OR_{12\backslash 12})$$

- More generally, we can show that $OR_{i_1 i_2, \backslash j j_2} = \exp(\beta_{i_1 j_1}^{XZ} + \beta_{i_2 j_2}^{XZ} - \beta_{i_1 j_2}^{XZ} - \beta_{i_2 j_1}^{XZ})$, where any parameters involving groups $i_1 = 1$, $i_2 = 1$, $j_1 = 1$, or $j_2 = 1$ are zero.

- Test for independence
  - Since the model without interaction parameters represents independence, we can test for independence with an LR test comparing two models:
    - $H_0 : X$ and Y are independent / $H_a : X$ and Y are not independent (or *are* associated)
    - Equivalently, $H_0 : \log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z$ / $H_a : \log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_{ij}^{XZ}$ for $i = 1, \ldots, I$; $j = 1, \ldots, J$
    - Equivalently, $H_0 : \beta_{ij}^{XZ} = 0$ for all $i = 1, \ldots, I$; $j = 1, \ldots, J$ vs. H_a : at least one $\beta_{ij}^{XZ} \neq 0$
  - Fit both models and compare residual deviances (or can just use `anova(H0,Ha)` or `Anova()` from `car`)

- Odds ratios now have meaning, not forced to be 1

<u>Example</u>: Energy Policy Opinions (Lecture 21 scripts.R, EnergyPolicy.csv)

Recall the energy policy example that we studied in Lecture 15 in the context of the multinomial distribution:

		Z =policy					
		conserve	oilgas	solar	nuclear	coal	Total
X =state	OILGAS	50	88	56	4	2	200
	COAL	59	20	52	3	66	200
	OTHER	161	40	188	5	6	400
	Total	270	148	296	12	74	800

Now we re-analyze the table in the context of a Poisson loglinear model. Let X represent the three types of state and Z represent the five policies. We will:

- fit the independence model and the saturated model,

- perform a likelihood ratio test for the interaction term in the saturated model,
- compare the test results to the LR test we performed assuming a product-multinomial model for the responses in each row,
- then use the model to compute a few odds ratios.

We have again reordered the factor levels to match the ordering in the table. Thus, the baseline row ($X = 1$) is `OILGAS` and the baseline column ($Z = 1$) is `conserve`. The independence model fits

$$\log(\mu) = \beta_0 + \beta_2^S x_2 + \beta_3^S x_3 + \beta_2^P z_2 + \beta_3^P z_3 + \beta_4^P z_4 + \beta_5^P z_5 = \beta_0 + \beta_i^S + \beta_j^P,$$

where I am again using variable initials as superscripts on the parameters (I can call them anything...). The model specifies an intercept, 2 `state` parameters and 4 `policy` parameters.

```
> mod.fit.ind <- glm(count ~ state + policy,
family=poisson(link="log"), data=set1)
> summary(mod.fit.ind)

Call: glm(formula = count ~ state + policy,
      family = poisson(link = "log"), data = set1)

Deviance Residuals:
   Min       1Q   Median       3Q      Max
-6.338   -2.884   -1.058    1.360    8.537

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    4.212e+00  8.633e-02  48.788  < 2e-16 ***
stateCOAL     -1.511e-15  1.000e-01   0.000    1.000
stateOTHER     6.931e-01  8.660e-02   8.004 1.21e-15 ***
policyoilgas  -6.012e-01  1.023e-01  -5.878 4.15e-09 ***
policysolar    9.194e-02  8.416e-02   1.092    0.275
policynuclear -3.114e+00  2.950e-01 -10.554  < 2e-16 ***
policycoal    -1.294e+00  1.312e-01  -9.864  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 790.77  on 14  degrees of freedom
Residual deviance: 249.17  on  8  degrees of freedom
AIC: 339.22

Number of Fisher Scoring iterations: 6
```

Sure enough, the model estimates exactly the parameters we expect to see. Not much to interpret here, because the main effect parameters mainly tell us about how the marginal totals differ from the first row or column. I will point out that the residual deviance is 249.17 on 8 DF ($8 = (I-1)(J-1)$,the number of missing interaction parameters).

The model containing the interaction terms comes next. We expect 8 additional parameters, one for each combination of `state` and `policy` beyond their first respective levels. Using the simpler notation, the model can be written as $\log(\mu_{ij}) = \beta_0 + \beta_i^S + \beta_j^P + \beta_{ij}^{SP}$, $i = 1, 2, 3$; $j = 1, 2, 3, 4, 5$.

```
> mod.fit.sat <- glm(count ~ state + policy + state:policy,
                     family=poisson(link="log"), data=set1)
> summary(mod.fit.sat)

Call: glm(formula = count ~ state + policy + state:policy,
          family = poisson(link = "log"), data = set1)

Deviance Residuals:
[1]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)                3.91202    0.14142  27.662  < 2e-16 ***
stateCOAL                  0.16551    0.19222   0.861  0.38920
stateOTHER                 1.16938    0.16190   7.223 5.09e-13 ***
policyoilgas               0.56531    0.17710   3.192  0.00141 **
policysolar                0.11333    0.19457   0.582  0.56026
policynuclear             -2.52573    0.51962  -4.861 1.17e-06 ***
policycoal                -3.21888    0.72111  -4.464 8.05e-06 ***
stateCOAL:policyoilgas    -1.64712    0.31355  -5.253 1.50e-07 ***
stateOTHER:policyoilgas   -1.95784    0.25015  -7.827 5.01e-15 ***
stateCOAL:policysolar     -0.23962    0.27210  -0.881  0.37851
stateOTHER:policysolar     0.04171    0.22223   0.188  0.85113
stateCOAL:policynuclear   -0.45320    0.78758  -0.575  0.56500
stateOTHER:policynuclear  -0.94624    0.69008  -1.371  0.17031
stateCOAL:policycoal       3.33099    0.74303   4.483 7.36e-06 ***
stateOTHER:policycoal     -0.07077    0.83239  -0.085  0.93225
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance:  7.9077e+02  on 14  degrees of freedom
Residual deviance: -3.0642e-14  on  0  degrees of freedom
AIC: 106.05
```

```
Number of Fisher Scoring iterations: 3
```

We see the additional parameter estimates we were expecting, each representing a different β_{ij}^{SP}, $i = 2, 3$; $j = 2, 3, 4, 5$. We also see that the residual deviance is now essentially zero (to 14 decimal places) on 0 degrees of freedom. That is the sure sign of a saturated model—one with as many parameters as modeled observations, leaving nothing "unexplained".

We can perform a LR test for the interaction term: $H_0 : \beta_{ij}^{SP} = 0$ for all i and j, vs. H_a :Not all $\beta_{ij}^{SP} = 0$. This is also equivalent to testing for independence in the contingency table, since all associations are contained in these interaction parameters.

```
> library(car)
> Anova(mod.fit.sat)
Analysis of Deviance Table (Type II tests)

Response: count
             LR Chisq Df Pr(>Chisq)
state           94.23  2  < 2.2e-16 ***
policy         447.38  4  < 2.2e-16 ***
state:policy   249.17  8  < 2.2e-16 ***
```

We see $-2\log(\Lambda) = 249.17$ and a tiny p-value on χ^2_8. Thus, at least one of the interaction parameters is needed in the model, and we do not have independence (or, we *do* have association) between people's energy policy preference and the energy production of the state they live in. If you go back to Lecture 15, you will see that this is *exactly* the same LR test statistic that was produced when we discussed the multinomial model for the table. This is no accident: mathematically, the likelihoods from the two models are equivalent. *Thus, it does not matter whether you use a multinomial model or a Poisson model as the basis for an analysis of a contingency table. They produce exactly the same results!*

The program for this example shows that the same LR test could be conducted by comparing the two fitted models using `anova()`. This will be useful to know when we start looking at loglinear models with 3 or more variables and more than one possible set of interaction parameters.

Next we start looking at means and various comparisons of them through mean ratios and odds ratios. The process is easy to do in `emmeans`. We simply use `emmeans()` to create a grid of means for each combination of `state` and `policy` using `specs=state+policy`. This automatically calculates the 15 estimated linear predictors (log-means) that we want to work with for any comparisons we want to make.

```
> emm1 = emmeans(mod.fit.sat, specs= ~state+policy)
> summary(emm1)
 state  policy   emmean     SE  df asymp.LCL asymp.UCL
 OILGAS conserve  3.912 0.1414 Inf     3.635      4.19
 COAL   conserve  4.078 0.1302 Inf     3.822      4.33
 OTHER  conserve  5.081 0.0788 Inf     4.927      5.24
```

```
 OILGAS oilgas    4.477 0.1066 Inf     4.268      4.69
 COAL   oilgas    2.996 0.2236 Inf     2.557      3.43
 OTHER  oilgas    3.689 0.1581 Inf     3.379      4.00
 OILGAS solar     4.025 0.1336 Inf     3.763      4.29
 COAL   solar     3.951 0.1387 Inf     3.679      4.22
 OTHER  solar     5.236 0.0729 Inf     5.093      5.38
 OILGAS nuclear   1.386 0.5000 Inf     0.406      2.37
 COAL   nuclear   1.099 0.5774 Inf    -0.033      2.23
 OTHER  nuclear   1.609 0.4472 Inf     0.733      2.49
 OILGAS coal      0.693 0.7071 Inf    -0.693      2.08
 COAL   coal      4.190 0.1231 Inf     3.948      4.43
 OTHER  coal      1.792 0.4082 Inf     0.992      2.59

Results are given on the log (not the response) scale.
Confidence level used: 0.95
```

Note that if we were to exponentiate each of these "emmeans," we would reconstruct the table counts perfectly, because the model is saturated. This is shown in the program.

We can use the `contrast()` function with various `method=` or `interaction=` arguments to make comparisons among these log means, and then express the results as mean ratios or odds ratios by using `type="response"`. Pairwise comparisons of two means at a time—differences of log-means exponentiated into ratios of means—can be performed using `method="pairwise"` or `"revpairwise"`, depending on the direction in which we want the comparisons done. For example, if we want to compare means for different policies within each state, we can do `method="pairwise", by="state"`. See the program for this analysis. It provides 30 mean ratios, which is a lot of ratios to sort through, but we look for patterns rather than try to interpret each ratio separately.

As an alternative to looking at all pairwise comparisons, **it might be more efficient and informative to compare each log-mean to the average of the other log means within that factor.** Note that this does all averaging in the log scale and exponentiates the result. [2] This is the `method="del.eff"` contrast. The results are below:

```
> confint(contrast(emm1, method="del.eff", by="state", type="response
state = OILGAS:
 contrast          ratio     SE  df asymp.LCL asymp.UCL
 conserve effect 3.5484 0.9301 Inf   1.80641     6.970
 oilgas effect   7.1932 1.7708 Inf   3.81528    13.562
 solar effect    4.0884 1.0558 Inf   2.10212     7.951
 nuclear effect  0.1510 0.0805 Inf   0.03823     0.596
 coal effect     0.0635 0.0457 Inf   0.00993     0.406
```

[2] It is NOT the same as comparing the mean for one one group against the average of means for the other groups. It *is* the same as comparing the mean for one group against the GEOMETRIC MEAN of the others, which is the k-th root of the product of k numbers.

```
state = COAL:
 contrast          ratio     SE  df asymp.LCL asymp.UCL
 conserve effect 2.7697 0.5747 Inf   1.62294     4.727
 oilgas effect   0.7164 0.1949 Inf   0.35543     1.444
 solar effect    2.3652 0.5028 Inf   1.36787     4.090
 nuclear effect  0.0669 0.0390 Inf   0.01490     0.300
 coal effect     3.1863 0.6481 Inf   1.88693     5.381

state = OTHER:
 contrast          ratio     SE  df asymp.LCL asymp.UCL
 conserve effect 7.3874 1.3012 Inf   4.69307    11.629
 oilgas effect   1.2958 0.2858 Inf   0.73422     2.287
 solar effect    8.9672 1.5580 Inf   5.73186    14.029
 nuclear effect  0.0963 0.0444 Inf   0.02936     0.316
 coal effect     0.1210 0.0515 Inf   0.04038     0.362

Confidence level used: 0.95
Conf-level adjustment: bonferroni method for 5 estimates
Intervals are back-transformed from the log scale
```

Looking at the `OILGAS` state, we see that a person is 7.2 times as likely to select `oilgas` policy as opposed to the average of the rest, with a 95% confidence interval (3.8, 13.6). In no other state is the support for `oilgas` policy more than the average of the other policies. Similarly, `COAL` state people express a preference for `coal` 3.2 times as often as the average of other policies (CI 1.9, 5.4), whereas in other states people prefer other policies to `coal` (the CIs lie entirely below 1). Interestingly, people in all states tend to prefer `conserve` or `solar` compared to other policies on average (and these CIs both exclude 1), while `nuclear` has little support.

Finally we look at some odds ratios. Since there are 3 ways to make pairs from 3 rows and 10 ways to make pairs from 5 columns, there are potentially $3 * 10 = 30$ different pairwise odds ratios that we could consider from each combination of pairs of rows and columns. Computing 30 ORs and trying to combine them into a sensible interpretation of the state-by-policy association seems a complex task. The output is shown in the program. Ordinarily, I'd rely on the subject-matter specialist to guide me as to which odds ratios make sense to examine. Since these data came from a book, there is no one to help me, so I will make up things that are interesting to *me*.

A very obvious thing to ask is whether the comparison of preference for oil-gas vs. coal policy depends on whether one is in an oil-gas state vs. a coal state. That is, we could compare the ratio of means for `oilgas/coal` policy in an `OILGAS` state to the same ratio in a `COAL` state. This is an odds ratio. We can read this one ratio out of the long list described in the previous paragraph, or we can compute this one contrast by itself by carefully selecting the groups for which we want to estimate linear predictors using the `at` argument in `emmeans`.[3] This analysis is done below:

[3]To request a single contrast in `emmeans`, we can specify coefficients similarly to what we do for `mcprofile`,

```
> emm.og = emmeans(mod.fit.sat, specs= ~state+policy,
+                  at=list(state=c("OILGAS", "COAL"),
+                          policy=c("oilgas","coal")))
> emm.og
 state  policy emmean    SE  df asymp.LCL asymp.UCL
 OILGAS oilgas  4.477 0.107 Inf     4.268      4.69
 COAL   oilgas  2.996 0.224 Inf     2.557      3.43
 OILGAS coal    0.693 0.707 Inf    -0.693      2.08
 COAL   coal    4.190 0.123 Inf     3.948      4.43

Results are given on the log (not the response) scale.
Confidence level used: 0.95
> OR.og = contrast(emm.og, interaction=c("pairwise", "pairwise"))
> confint(OR.og, type="response")
 state_pairwise policy_pairwise ratio  SE  df asymp.LCL asymp.UCL
 OILGAS / COAL  oilgas / coal     145 110 Inf      32.8
643

Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

The mean ratio (or odds!) of preference for oil/gas policy over coal policy are estimated to be 145 times as high among people in oil/gas producing states compared to those in coal-producing states, with a 95% Wald confidence interval from 33–643. Clearly, the state's energy production is strongly associated its residents' policy opinions. Note that we do not establish causality here. While it is tempting to conclude that the state's economic benefit drives people to want to defend or maintain that benefit, it is at least slightly conceivable that people's opinions help to direct the kind of energy that a state pursues, or that people move to states that support their interests.

To show the potential flexibility that the loglinear model affords, let's consider a different, more complex question. Let's try to answer the question: Is the preference for conservation vs. alternative energy development (solar or nuclear) the same among people in "neutral" states as it is among energy-producing states? Formulating this in terms of the available categories, is the ratio of `conserve` vs. `(solar, nuclear)` the same in `OTHER` states as it is in `(OILGAS,COAL)` states?

---

but it is easier. We need to know which *means* to compare, rather than which *parameters*. Looking at the ordering of the means in the `emm1` object, we want to make a ratio of μ_4/μ_{13} vs. μ_5/μ_{14}. In terms of log-means we have

$$\log(OR) = [\log(\mu_4) - \log(\mu_{13})] - [\log(\mu_5) - \log(\mu_{14})].$$

This this corresponds to these coefficients on the 15 means:

$$0, 0, 0/1, -1, 0/0, 0, 0/0, 0, 0/-1, 1, 0$$

where I am using the slashes to separate out the groups of 3 states for each policy as they are ordered in `emm1`.

```
> emm.alt = emmeans(mod.fit.sat, specs= ~state+policy,
+                   at=list(policy=c("conserve", "solar", "nuclear")))
> emm.alt
 state  policy   emmean     SE  df asymp.LCL asymp.UCL
 OILGAS conserve   3.91 0.1414 Inf     3.635      4.19
 COAL   conserve   4.08 0.1302 Inf     3.822      4.33
 OTHER  conserve   5.08 0.0788 Inf     4.927      5.24
 OILGAS solar      4.03 0.1336 Inf     3.763      4.29
 COAL   solar      3.95 0.1387 Inf     3.679      4.22
 OTHER  solar      5.24 0.0729 Inf     5.093      5.38
 OILGAS nuclear    1.39 0.5000 Inf     0.406      2.37
 COAL   nuclear    1.10 0.5774 Inf    -0.033      2.23
 OTHER  nuclear    1.61 0.4472 Inf     0.733      2.49

Results are given on the log (not the response) scale.
Confidence level used: 0.95
> OR.alt = contrast(emm.alt, interaction=c("del.eff", "del.eff"))
> # See "OTHER effect" + "conserve effect" to compare OTHER states
> #   to energy producers and conserve policy to alt fuels.
> confint(OR.alt, type="response")
 state_del.eff policy_del.eff  ratio    SE  df asymp.LCL asymp.UCL
 OILGAS effect conserve effect 0.671 0.240 Inf     0.333
1.35
 COAL effect   conserve effect 1.128 0.424 Inf     0.540
2.36
 OTHER effect  conserve effect 1.322 0.429 Inf     0.699
2.50
 OILGAS effect solar effect    0.778 0.276 Inf     0.388
1.56
 COAL effect   solar effect    0.763 0.288 Inf     0.364
1.60
 OTHER effect  solar effect    1.684 0.546 Inf     0.893
3.18
 OILGAS effect nuclear effect  1.916 1.205 Inf     0.558
6.57
 COAL effect   nuclear effect  1.162 0.786 Inf     0.308
4.38
 OTHER effect  nuclear effect  0.449 0.267 Inf     0.140
1.44

Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

The estimated OR is listed as `OTHER effect` , `conserve effect`, and the estimate is 1.32, indicating that the estimated ratio of means (or the odds) of a person preferring conservation

over alternative energy production is 32% higher for people in non-energy producing states than for those in energy-producing states. The 95% Wald confidence interval is 0.70 to 2.5, so we do not have sufficient evidence to declare that these two mean ratios or odds are different from each other. In other words, there is no clear difference in the comparison of conservation vs. alternative energy for people in energy or non-energy states.

While this example is a little complicated, it shows that we can be creative in the definition of odds ratios and are not limited to just selecting two rows and two columns. We can combine rows and columns to create different mean counts as interests arise. In the appendix I show how to do LR inference using `mcprofile`. The process is a bit more complicated, because we have to carry the algebraic comparisons out one step further, replacing the log-means with their corresponding model parameters, in order to determine the required contrast coefficients.

---

3 Notes

- ANY logistic regression model with a categorical explanatory variable can be rewritten as a Poisson loglinear model.
  - The problem amounts to modeling counts an $I \times 2$ table
  - Loglinear model will have more complicated structure.
  - But loglinear model is not limited to binary responses, where logistic regression is
  - And loglinear model can be used when there is no natural "response" variable
    - Many surveys are like this
    - See political ideology example in next lecture
- The loglinear model can be extended to allow <u>more than two variables</u> (next lecture) and can accommodate ordinal variables (the lecture after that)
- The Poisson regression model can also allow numerical and categorical variables together in the same model

4 Conclusions: What to learn

1. How to create loglinear (Poisson regression) models out of two categorical variables
2. How test associations between variables and what the association means
3. How to form, estimate, and interpret odds ratios from cells in a table
   (a) `emmeans` is your friend here—saves having to manipulate parameters in a linear predictor.

5 Appendix 1: Complicated Analysis in `mcprofile`

To express log means as parameters, we need to return to our μ_{ij} notation, where i is the state and j is the policy. The first comparison is the ratio of means for `oilgas/coal` policy in an `OILGAS` state to the same ratio in a `COAL` state. Numerically, this is μ_{12}/μ_{15} vs. μ_{22}/μ_{25} or $\log(OR_{12\backslash 25}) = [\log(\mu_{12}) - \log(\mu_{15})] - [\log(\mu_{22}) - \log(\mu_{25})] = \beta_{12}^{SP} - \beta_{15}^{SP} - \beta_{22}^{SP} + \beta_{25}^{SP}$. Since the first two of these parameters are for the first level of a variable, they are automatically 0. Thus, we need to have a -1 for `stateCOAL:policyoilgas` and a $+1$ for `stateCOAL:policycoal`. Considering the order of the parameters in the `summary()` output, we need a coefficient vector that has seven 0's, a -1, five more 0's, a 1, and one last 0. The results are below:

```
> K1 <- matrix(c(rep(0, 7), -1, rep(0,5), 1, 0), nrow=1)
> OR1 <- mcprofile(object=mod.fit.sat, CM=K1)
> exp(confint(OR1))

   mcprofile - Confidence Intervals

level:           0.95
adjustment:      single-step

   Estimate lower upper
C1      145  40.8   931
```

The estimated OR is identical to what we found from `emmeans`, as it should always be. The CI is somewhat different, though, shifted to higher lower and upper endpoints. Both are so far from 1 that there is no question about the large magnitude of this OR.

The next question was: Is the preference for conservation vs. alternative energy development (solar or nuclear) the same among people in "neutral" states as it is among energy-producing states? Formulating this in terms of the available categories, is the ratio of `conserve` vs. `(solar, nuclear)` the same in `OTHER` states as it is in `(OILGAS,COAL)` states? Numerically, this is level 1 vs. levels (3,4) of `policy` at `state` level 3 compared to the same comparison at state levels (1,2). The contrast in terms of state and policy indexes is

$$[\log(\mu_{31})-(\log(\mu_{11})+\log(\mu_{21}))/2]-[(\log(\mu_{33})+\log(\mu_{34}))/2-(\log(\mu_{13})+\log(\mu_{14})+\log(\mu_{23})+\log(\mu_{24}))/4].$$

As an exercise, ***work out*** *what the coefficients should be for this comparison.* It's just a matter of substituting parameters for the log means and then doing a lot of tedious algebra. I will give you the final answer:

$$0, 0, 0, 0, 0, 0, 0, 0, 0, .25, -.5, .25, -.5, 0, 0$$

```
> K2 <- matrix(c(0,0,0,0,0,0,0,0,0,.25,-.5,.25,-.5,0,0), nrow=1)
> OR2 <- mcprofile(object=mod.fit.sat, CM=K2)
> exp(confint(OR2))

   mcprofile - Confidence Intervals

level:           0.95
adjustment:      single-step

   Estimate lower upper
C1     1.32 0.701  2.57
```

This is very similar to the results from above.

Now you know why `emmeans` is your friend...

6 Exercises (due when announced)

1. In the 2×2 example on page 6, when I write the model, $\log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_{ij}^{XZ}$, for $i = 1, 2$ and $j = 1, 2$, what I really mean is

$$\begin{array}{l} \log(\mu_{11}) = \beta_0 + \beta_1^X + \beta_1^Z + \beta_{11}^{XZ} \\ \log(\mu_{12}) = \beta_0 + \beta_1^X + \beta_2^Z + \beta_{12}^{XZ} \\ \log(\mu_{21}) = \beta_0 + \beta_2^X + \beta_1^Z + \beta_{21}^{XZ} \\ \log(\mu_{22}) = \beta_0 + \beta_2^X + \beta_2^Z + \beta_{22}^{XZ} \end{array}.$$

   Explain why this model is exactly equivalent to the one we derived from the table, $\log(\mu_{ij}) = \beta_0 + \beta_2^X x_2 + \beta_2^Z z_2 + \beta_{22}^{XZ} x_2 z_2$. (The point here is for you to fully appreciate that the simplified version of the model is exactly the same as what you get when you write out the full regression model with indicators. We will find this simplified notation to be *much* easier to work with as we move forward.)

2. Refer to the data in Table 3.5 of B&L, page 236. The description of these data is on the previous page, in Exercise 22 from Chapter 3. This is another data set that I have worked on. (The researcher who provided these data to me, Dusti Becker, was once Dean of Natural Resources & Applied Science at Solomon Islands National University!)

   (a) We use "independence" to mean different things in different contexts.

      i. Suppose we want to model the counts on the table using a Poisson regression model. We assume that the random variables representing the counts in each cell of the table are independent of each other. What does this form of "independence" mean in the context of this problem?

      ii. We can test for "independence" between the row and column variables in the table. This is a completely different notion from the "independence" of the Poisson random variables. What does "independence" between degradation and guild mean in the context of the problem?

   (b) Fit the independence loglinear model. You don't have to write out the estimated model. Instead, just report the residual deviance and corresponding degrees of freedom.

   (c) Fit the saturated model. Report the LR test for independence and interpret the result. Include all parts of the test.

   (d) Estimate the ratio of mean counts under high vs. low degradation for each guild. Find the corresponding confidence intervals.

      i. Do any guilds seem to have lower ratios than others?

      ii. Explain why this ratio measures the potential adverse effect of habitat degradation on a given guild, and why ratios that are far below 1 imply greater potential impact.

   (e) Since the habitat in question is specifically forest, one might expect that birds that require forest stands would be more negatively impacted by habitat degradation than others. That is, they would be less abundant where there is high amounts

of degradation than there would be otherwise. To test this theory, compare the high:low mean ratio computed in the previous part for `Wood-Arth` to the ratio for each other guild in a separate comparison (there will be 6 comparisons). Compute the odds ratios and the corresponding confidence intervals.

i. Provide a table of the estimated odds ratios and the corresponding confidence intervals.

ii. Use this table to explain whether `Wood-Arth` guild does experience a greater reduction in abundance in high vs. low degradation than other guilds.

NOT COLLECTED:

1. As suggested in the appendix, manipulate the log-means for the odds ratio from the last example, to show that the coefficients on the parameters that were given in the example are correct.